Simulations and Preliminary studies of radars using illuminators of opportunity

A Thesis Submitted
In Partial Fulfillment of the Requirements
for the Degree of ***M.Tech***

Anunnya Gudhenia

Roll No.: 241310002

to the

Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE)

Indian Institute of Technology Kanpur.

May, 2026

Certificate

It is certified that the work contained in the thesis titled **"Simulations and Preliminary studies of radars using illuminators of Opportunity"**, by **"Anunnya Gudhenia"** (**Roll No. 241310002**) has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Prof. Mugundhan Vijayaraghavan
Dept. of Space, Planetary & Astronomical
Sciences & Engineering (SPASE),
Indian Institute of Technology Kanpur.

May, 2026

Declaration

This is to certify that the thesis titled **"Simulations and Preliminary studies of radars using illuminators of opportunity"** has been authored by me. It presents the research conducted by me under the supervision of **Prof. Mugundhan Vijayaraghavan**.

To the best of my knowledge, it is an original work, both in terms of research content and narrative, and has not been submitted elsewhere, in part or in full, for a degree. Further, due credit has been attributed to the relevant state-of-the-art and collaborations with appropriate citations and acknowledgments, in line with established norms and practices.

Anunnya Gudhenia
Roll No. 241310002
Dept. of Space, Planetary & Astronomical
Sciences & Engineering (SPASE)
Indian Institute of Technology Kanpur,
Kanpur, 208016, India.

Synopsis

Anunnya Gudhenia: Your Name **Roll No**: 241310002
Degree for which submitted: M.Tech **Department**: Dept. of Space, Planetary & AstronomicalSciences & Engineering (SPASE) **Thesis Title**: Simulations and Preliminary studies of radars using illuminators of opportunity"
Thesis Supervisor: Prof. Mugundhan Vijayaraghavan
Month and Year of Thesis Submission: May, 2026

This thesis presents simulations and preliminary experimental studies of passive bistatic and multistatic radar systems using illuminators of opportunity available within India's existing scientific infrastructure. FM broadcast stations in Kanpur were first evaluated as illuminators of opportunity. A three-element Yagi-Uda antenna was designed using 4NEC2 simulation software, fabricated, and experimentally validated, achieving a gain of 7.02 dBi and a front-to-back ratio of 33.3 dB. Strong direct-path interference from FM transmitters limited practical target detection, motivating the use of the ARIES ST Radar at Nainital (206.5 MHz, 235 kW) as a more suitable illuminator of opportunity. A seven-element Yagi-Uda antenna optimised for 206.5 MHz was subsequently designed (gain 10.11 dBi, front-to-back ratio 56.39 dB, HPBW 48 and fabricated. A 22-hour continuous data acquisition experiment was performed using the antenna, a low-noise amplifier, and an RTL-SDR receiver. A total of 800,469 power spectrum frames were recorded. The ARIES carrier was detected across the 337 km bistatic baseline between Nainital and IIT Kanpur. A drift scan of the dataset confirmed a Galactic centre transit at LST $\approx$ 17.30 h, validating the antenna beam and data acquisition system. Bistatic and multistatic radar performance simulations characterised system SNR as a function of pulse repetition frequency, pulse width, target radar cross section, and number of receive array elements. Multistatic geometry simulations demonstrated that a second receiver resolves the position ambiguity inherent in a single bistatic measurement through the intersection of constant bistatic range ellipses. The work establishes a proof-of-concept for a low-cost passive radar network using Indian scientific infrastructure as illuminators of opportunity.

Acknowledgements

would like to express my sincere gratitude to my guide, Dr. Mugundhan Vijayaraghavan, for his constant guidance, encouragement, and valuable suggestions throughout the course of this work. His insights and support helped me understand the practical and theoretical aspects of passive radar systems, Software Defined Radio, and antenna array processing.

I would like to thank Ms. Jayasandhya Meenakshinathan, Research Establishment Officer at SPASE department. She has helped me throughout my simulations and experiments. I would like to extend my gratitude to Mr. Kundan Kharwar, Jr. Technician in SPASE department. He helped me to build both of the antennas,and during the experiments as well. I am thankful to my friend Mr. Ashwin Raju Kharat, and Mr. Shubham Khose for helping out during measurements and readings at various stages of thesis. I am grateful to Ms. Shreya Chaturvedi for her support and encouragement during various stages of the my thesis.

I would also like to acknowledge the developers and contributors of open-source tools and libraries such as GNU Radio, RTL-SDR, Python scientific libraries, and 4NEC2 software, which played an important role in the completion of this work.

Finally, I express my heartfelt gratitude to my parents and my sister for their continuous support, motivation, and encouragement throughout my academic journey.

Anunnya Gudhenia

Contents

List of Figures

List of Tables

Abbreviations

ARIES	**A**ryabhatta **R**esearch **I**nstitute of**O**bservational **S**ciences
BIRALES	**BI**static **RA**dar for **LE**o **S**urveillance
BIRALET	**BI**static **RA**dar for **LE**o **T**racking
DAB	**D**igital **A**udio **B**roadcasting
DVB-T	**D**igital **V**ideo **B**roadcasting **T**errestrial
EIRP	**E**ffective **I**sotropic **R**adiated**P**ower
FFT	**F**ast **F**ourier **T**ransform
FM	**F**requency **M**odulation
HPBW	**H**alf-**P**ower **B**eam**W**idth
IOO	**I**lluminator **O**f **O**pportunity
LEO	**L**ow **E**arth **O**rbit
LNA	**L**ow **N**oise **A**mplifier
LOFAR	**LO**w **F**requency **AR**ray
LST	**L**ocal **S**idereal **T**ime
MWA	**M**urchison **W**idefield **A**rray
NEC	**N**umerical **E**lectromagnetics **C**ode
NF	**N**oise **F**igure
PCL	**P**assive **C**oherent **L**ocation
PDF	**P**robability **D**ensity **F**unction
PRF	**P**ulse **R**epetition **F**requency
RADAR	**RA**dio **D**etection **A**nd **R**anging
RCS	**R**adar **C**ross **S**ection
RTL-SDR	**R**ealtek **T**elevision **L**ogic **S**oftware **D**efined **R**adio
SNR	**S**ignal-to-**N**oise **R**atio
SRT	**S**ardinia **R**adio **T**elescope
SWR	**S**tanding **W**ave **R**atio
TOA	**T**ime **O**f **A**rrival

VHF	**V**ery **H**igh **F**requency
VNA	**V**ector **N**etwork **A**nalyser

Symbols

A_e	Effective Aperture of Antenna
B	Bandwidth
β	Bistatic Angle
c	Speed of Light
D	Directivity
D_0	Maximum Directivity
d	Inter-element Spacing of antennas
E	Electric Field
f	Frequency
f_d	Doppler Frequency Shift
f_s	Sampling Rate
F	Noise Factor
F_{sys}	System Noise Factor
G	Antenna Gain
G_r	Receiver Antenna Gain
G_t	Transmitter Antenna Gain
k	Boltzmann's Constant
λ	Wavelength
N	Number of Array Elements
NF	Noise Figure
P_{EIRP}	Effective Isotropic Radiated Power
P_n	Noise Power
P_r	Received Power
P_{rad}	Total Radiated Power
P_{sig}	Received Signal Power
P_t	Transmitted Power

R	Range
R_R	Target-to-Receiver Distance
R_T	Transmitter-to-Target Distance
$\hat{R}_R$	Unit Vector from Target to Receiver
$\hat{R}_T$	Unit Vector from Target to Transmitter
S	Power Density
σ	Radar Cross Section
$\sigma_{b,\min}$	Minimum Detectable Bistatic RCS
SNR	Signal-to-Noise Ratio
T	Temperature
τ	Pulse Width / Propagation Delay / Integration Time
U	Radiation Intensity
v	Target Speed
$\mathbf{w}$	Weight Vector
ψ	Phase Shift between Array Elements
$\mathbf{a}(\theta)$	Steering Vector
AF	Array Factor

Dedicated to my beloved parents ans sister.

Chapter 1

Introduction

1.1 Introduction to Radar Systems

The term RADAR stands for Radio Detection and Ranging [8]. Radar systems use electromagnetic waves to detect objects and determine their positions, distances, and motions. Today, radar has become an essential technology in many areas such as military surveillance [9], weather forecasting [8], remote sensing [10], and space research [11] [12].

Radar technology was originally developed during the early twentieth century, mainly for the detection of aircraft and for defence applications during wartime [8]. Since then, continuous developments in electronics, antennas, and signal processing have made radar systems more accurate, reliable, and versatile. Modern radar systems are capable of operating in different environments and under weather conditions where normal optical systems or human vision don't work properly [1].

Radar transmits electromagnetic energy through an antenna. This radiated energy gets scattered by the target. The scattered energy, also known as echo, gets reflected towards the receiver. This signal can be used to find out useful information about the target [8], [1]. This means a simple radar consists of a transmitter, a transmitting antenna, a receiving antenna and a signal processing unit.

Radars are able to find the target distance by measuring the time delay between the transmission and receiving of the signal after it gets reflected from the target. If it is a moving target, it will produce frequency shifts. So, The Doppler effect can be used to find the velocity of the moving target as well [8].The Doppler effect has been explained in section(2.2.3). Over the years, radar systems have evolved from conventional monostatic systems to more advanced bistatic, multistatic, and passive radar configurations.

Passive Radars are the type of radars which do not need a dedicated transmitter. They use existing broadcast signals or communication signals. This not only makes them cost-effective but also suitable for covert operations. This has led to the rise in popularity of Passive Radar Systems [13]

1.2 Basic Parts of a Radar

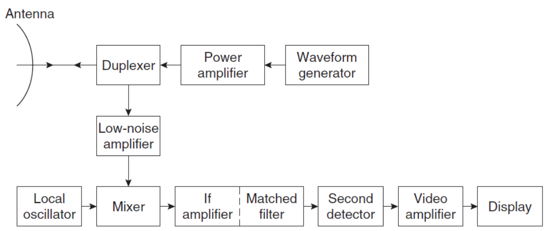

FIGURE 1.1: Block diagram of a simple radar employing a power amplifier as the transmitter in the upper portion of the figure and a superheterodyne receiver in the lower portion of the figure. [1]

A radar system has different sections that work together to locate and detect targets [14]. The transmitter generates high-power electromagnetic signals that are transmitted with the help of antenna(s). On the basis of transmission type, radar systems can be categorized as Continuous wave Radars or Pulsed Radars. Pulse

Radars are most common as they allow transmission and reception of signals through the same antenna via a duplexer. [1].

The duplexer behaves like a switch between the transmitter and the receiver. During transmission, it protects the sensitive receiver circuitry from the high transmitted power. After transmission, it connects the antenna to the receiver so that the reflected echoes can be received safely [15]. The antenna transmits electromagnetic energy towards the target and also receives the reflected signals being scattered from the object. Radar antennas are generally designed to produce narrow beams so that the transmitted energy is concentrated in a particular direction. This improves detection capability and helps in determining the direction of the target more accurately [16], [17].

The reflected signals that reach the receiver are generally very weak.They need to be amplified to make them useful for further processing. A good receiver is highly sensitive and has low internal noise [15], [10].

After the signal is received, the signal processor extracts useful information from it. Filtering, Doppler processing, clutter suppression, and target detection happen at this stage. Signal processing improves the quality of the received data and helps separate actual target echoes from unwanted signals caused by noise or surrounding objects [1]. In Passive Radar Systems signal processing also includes Direct Signal Interference(DSI), Cancellation and Cross-Ambiguity Function(CAF) [13], [10]

Once the target information is extracted, it can either be displayed to an operator or used in automatic tracking systems. Modern radar systems are capable of continuously tracking targets and estimating parameters such as range, direction, and velocity [1].

The radar control system coordinates the operation of all radar components. It provides timing and synchronization signals required for proper transmission, reception, and processing, by setting parameters like Pulse Repetition Frequency, waveform type and receiver gain [1], [15]. In practice, radar performance also depends on external factors such as atmospheric conditions, unwanted reflected signals known as clutter, interference, and the characteristics of the target itself [15]. These factors can affect the accuracy and reliability of detection and tracking [1].

1.3 Information available from a Radar

As discussed in previous sections, Radars have the capability to not only detect but also provide useful information about the target such as its distance, direction, speed, movement and even the size and shape. It depends on the radar type as well as the signal processing techniques that are used. [1], [18].

The range of the target is calculated by measuring the time difference between the transmitted wave and the received echo. Since radio waves travel at the speed of light, even small time differences can be used to determine distance with high accuracy [19].

When the target is in motion, the frequency of the received echo changes slightly due to the Doppler effect. Radars measure this frequency shift, to estimate its relative speed [15].

A directional antenna can determine the the angular position of the target. The direction where the received signal strength becomes maximum gives the approximate location of the target in terms of azimuth and elevation angles [16].

Higher bandwidth signals provide better range resolution and improves the accuracy of measurements, allowing nearby targets to be separated more clearly. Advanced imaging techniques such as Synthetic Aperture Radar (SAR) and Inverse Synthetic Aperture Radar (ISAR) are capable of producing more detailed target images. This can help find the target size and shape [1], [10]. Modern radar systems often use multiple operating frequencies to improve performance. Techniques such as frequency agility and frequency diversity help reduce interference, improve detection probability, and make the radar less vulnerable to jamming [1], [15].

1.4 History of Radar

In the early twentieth century, several scientists and engineers explored the possibility of using radio waves to detect distant objects. In 1900, Nikola Tesla discussed the concept of electromagnetic detection and velocity measurement [14]. A few years later, during 1903 and 1904 the German engineer Christian Hülsmeyer carried out experiments on detecting ships using reflected radio waves and received patents for an obstacle detection device [8]. However, due to limited available technology, the system could not achieve large detection ranges.

In 1922, Guglielmo Marconi suggested that short radio waves could be used for detection purposes [8]. Around the same time, A. H. Taylor and L. C. Young of the U.S. Naval Research Laboratory demonstrated the detection of ships using reflected radio signals. At this time, the systems were only capable of detecting a target. The study by Tyalor and Young was inherently bistatic in nature as the receiver was separated from the transmitter [14].

The first successful aircraft detection using radio waves was carried out in 1930 by L. A. Hyland at the Naval Research Laboratory [8], [14]. Even though, the observation was accidental, it still encouraged further investigation into radio-based detection systems. The development of Pulse Radars, in 1930s, was due to the inability of continuous wave radar systems to measure target range accurately. Pulse radar allowed target range to be measured by calculating the time delay between transmission and reception of the reflected signal [8]. In the United States, R. M. Page and his team at the Naval Research Laboratory successfully demonstrated pulse radar operation in 1936 [14].

Radar development progressed rapidly during World War II. Several countries including the United States, Britain, Germany, France, Russia, Italy, and Japan carried out independent radar research programs [14]. In Britain, Sir Robert Watson-Watt led the development of the Chain Home early warning radar network [8].

Initially, radar technology was developed mainly for military applications such as surveillance, navigation, target tracking, and fire-control systems [14]. After World War II, radar systems gradually found applications in civilian fields as well.

1980s and 1990s saw a keen interest in Passive Cherent Location(PCL) systems using broadcast transmitters as Illuminators of Opportunity. [13], [18]. In 1998, Lockheed Martin demonstrated the Silent Sentry passive coherent location system, which used FM radio and analogue television signals as illuminators of opportunity for wide-area air surveillance [13], [18].

More recently, passive radar has been applied to space situational awareness. The Murchison Widefield Array (MWA) in Australia demonstrated passive radar detection of LEO space objects using existing VHF transmitters as IOOs [20]. Similarly, the LOFAR radio telescope array in Europe has been used as a passive receiver in a multistatic configuration for tracking LEO debris [12]. Parameter estimation techniques for LEO debris using passive radar have also been developed and validated [21]. In-orbit passive bistatic radar using CubeSats has been proposed as a novel low-cost approach to debris detection from orbit [22].

1.5 Classification of Radars

Radar systems can be classified based on the relative placement of the transmitter and receiver. Depending on the system geometry, radar systems are generally categorized as monostatic, bistatic, and multistatic radar systems.

1.5.1 Monostatic Radar

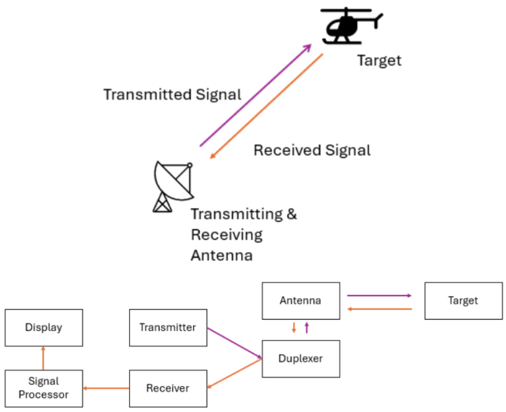

FIGURE 1.2: Monostatic Radar System Block Diagram

Fig (1.2) shows a monostatic radar system is the most commonly used radar configuration in which the transmitter and receiver are located at the same position and generally share a common antenna through a duplexer. In this type of radar system, electromagnetic energy is transmitted toward a target, and the reflected echo from the target is received by the same radar system [14], [23].The fraction of incident electromagnetic energy scattered back toward the radar by a target is characterised by the radar cross section (RCS). The RCS depends on the shape, size, material, surface finish, and orientation of the target relative to the radar, as well as the operating frequency [1], [15].

Monostatic radar systems are widely used in Air surveillance radar, Weather radar, Marine radar, Fire-control radar, Automotive radar, Air traffic control systems [10],

[15]. Monostatic radar systems offer several important advantages. They provide full control over the transmitted frequency, waveform, bandwidth, pulse shape, modulation, and power which simplifies matched filtering and range Doppler Processing [10], [15]. The co-location of transmitter and receiver simplifies synchronization and calibration and the technology is well-developed with decades of operational experience [8]. However, there are some limitations of monostatic radar. Since the system continuously radiates electromagnetic energy, it is detectable and vulnerable to anti-radiation missiles and jamming [9], [13]. Stealth technology specifically targets the monostatic geometry. These limitations have motivated significant research into bistatic and passive radar configurations that do not rely on a dedicated co-located transmitter [18], [24].

1.5.2 Bistatic Radar

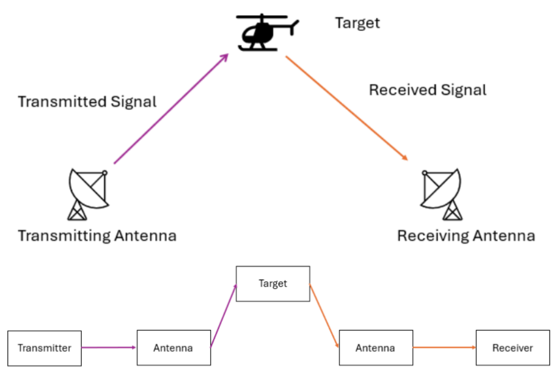

FIGURE 1.3: Bistatic Radar System Block Diagram

Fig (1.3) shows an active bistatic radar system, the transmitter and receiver are placed at different locations and are separated by a considerable distance. Unlike monostatic radar systems, where the same antenna is used for transmission and reception, bistatic radar uses separate transmitting and receiving stations [13].

The transmitted electromagnetic wave travels from the transmitter to the target, gets scattered and reaches the receiver at a different location. Because the transmitter, target, and receiver are located at different positions, the overall geometry of the system becomes more complex. This arrangement is commonly referred to as bistatic geometry [15], [19].

A bistatic radar system generally includes transmitter, transmitting antenna, receiving antenna, receiver, signal processing unit, synchronization and control system [13]. A Bistatic Radar which uses external transmitters like FM radio stations, TV Broadcast tower, satellite transmission or communication signals are known as Passive Radars [24], [18]. Unlike monostatic radar, the received power in bistatic radar depends on two separate propagation distances instead of a single round-trip distance [13].

There has been significant development in Bistatic and Passive Radars experiments. FM broadcast stations have been used as illuminators of opportunity to detect and track aircraft at ranges exceeding 150 km [19], [25]. Digital Video Broadcasting-Terrestrial (DVB-T) transmitters have been exploited as wideband IOOs due to their OFDM signal structure. This provides improved range Doppler resolution compared to FM [26], [27].

Cristallini et al., used transmissions from satellite as illuminators of opportunity for multiband passive radar imaging, which provided a wide-area coverage [28].

Fig (1.4) The Italian BIRALET system is a bi-static radar developed for space debris detection and tracking in Low Earth Orbit. It uses the transmitter operating at 410–415 MHz as the transmitter and the Sardinia Radio Telescope (SRT) as the receiver. The system is designed to perform range and range-rate measurements of space objects with high accuracy. [2]

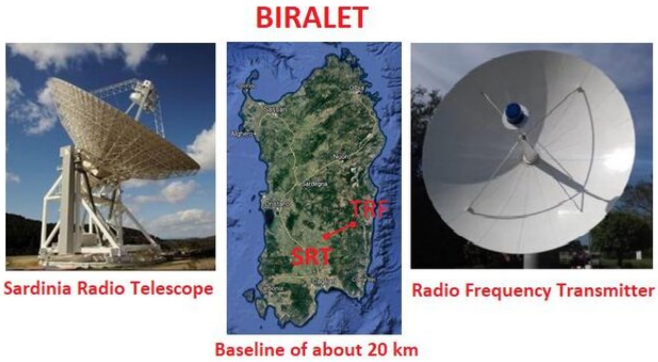

FIGURE 1.4: The BIRALET radar [2]

Fig (1.5) shows a part of the Murchison Widefield Array in Western Australia demonstrated passive radar detection and tracking of LEO space objects using existing VHF broadcast transmitters as IOOs and the MWA radio telescope array as the passive receiver [20]. The MWA operates at 70–300 MHz and provides an enormous effective collecting area. The remote location of the MWA in the Western Australian outback minimises radio frequency interference [29], [20].

FIGURE 1.5: A view of part of the Murchison Widefield Array [3]

Bistatic and passive radar systems have received significant research attention because of their low cost, covert operations, and ability to utilise existing infrastructure for target detection and surveillance [20], [21]. Bistatic radar systems are widely used in military surveillance, missile guidance, passive coherent location (PCL) systems, target classification and tracking, space and planetary radar, and passive radar networks [13], [24].

Bistatic radar systems also offer reduced vulnerability to jamming since the receiver does not radiate and its location may be unknown to an adversary [9], [18]. The Bistatic Geometry provides better spatial coverage can be achieved since the receiver can be deployed independently of the transmitter [1], [13].

However, there are some complexities that arise with bistatic geometry. Accurate synchronisation between the spatially separated transmitter and receiver is essential, particularly when the transmitter is a noncooperative IOO whose exact waveform parameters may not be fully known [13], [19]. Signal processing is also more complicated because the transmitted waveform must first be reconstructed from the reference channel before the cross-ambiguity function can be computed [30], [13], [31]. Target localisation is more complex because the bistatic range measurement places the target on an ellipse rather than at a unique point, requiring additional information such as angle of arrival or a second bistatic measurement to fully resolve the target position [12], [20]. Target Localisation can be solved with introducing more transmitters in receivers in different locations.

1.5.3 Multistatic Radar

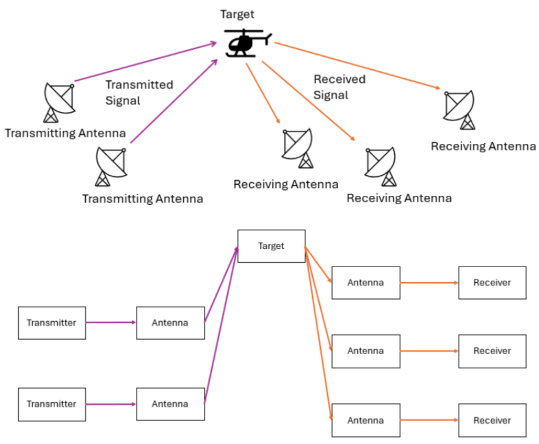

FIGURE 1.6: Multitatic Radar System Block Diagram

Fig (1.6) shows a multistatic radar system. They use multiple receivers and/or transmitters placed at different geographical locations [8], [13]. In these systems, the target is observed from several directions instead of a single location. Because of this distributed arrangement, multistatic radar systems can provide better coverage and improved detection capability compared to conventional monostatic radar systems [13], [18]. In a multistatic radar configuration, several receiving stations may receive reflections from the same target simultaneously.As in bistatic radar the transmitters may either be dedicated radar transmitters or external illuminators of opportunity such as FM radio stations, television broadcasts, communication towers, or satellite signals [18], [24]. A multistatic radar system generally consists of multiple transmitting stations, receiving stations, or both. One of the main advantages of multistatic radar is spatial diversity. Since the target is viewed from multiple directions, the probability of detection improves.This becomes particularly useful in the detection of stealth targets [1], [13]. Each transmitter-receiver pair in a multistatic system forms an independent bistatic path, and the intersection of the corresponding constant bistatic range ellipses from multiple pairs localises the target to a point, resolving the position ambiguity inherent in any single bistatic measurement [12], [20]. Using measurements from several receivers also increases reliability and reduces the effect of interference or signal blockage at any particular location [18], [24].

Compared to monostatic and bistatic radar systems, multistatic radar geometry is more complex because each transmitter–receiver pair forms an independent bistatic path. As a result, accurate synchronization between all nodes becomes very important. Signal processing is also more demanding because information from different receiving stations must be processed together [13], [18].

Multistatic radar systems are used in air defence, passive radar networks, distributed sensing, wide-area surveillance, border monitoring, and stealth target detection [13], [9].

The most recent and technically advanced multistatic passive radar demonstration was done using a scientific radio telescope is the LOFAR-based system described by Jędrzejewski et al. [12]. This system uses terrestrial VHF digital radio transmitters as IOOs and the LOFAR (Low-Frequency Array) radio telescope array in Poland as the passive receiver in a multistatic configuration. By using multiple transmitter and receiver pairs, the system resolves the bistatic position ambiguity, providing three-dimensional target localisation of LEO space objects.

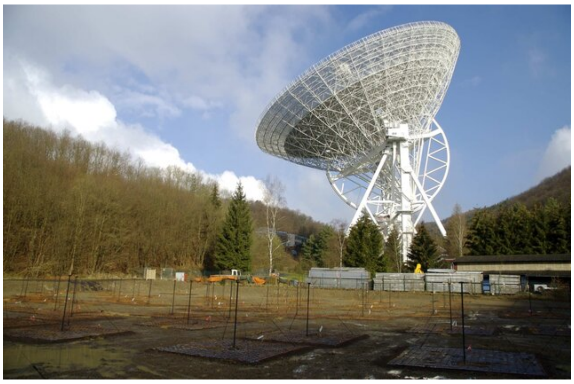

FIGURE 1.7: Low Frequency Array Radio Telescope [4]

With recent developments in digital signal processing, software-defined radio platforms, and networking technologies, multistatic radar has become an important research area in modern radar and passive surveillance [32], [33].

1.6 Comparison of Active and Passive Radar Systems

TABLE 1.1: Comparison of Active and Passive Radar Systems

Aspect	**Active Radar**	**Passive Radar**
System Architecture	Has a dedicated transmitter and receiver, often co-located. The transmitted signal is known and fully controlled [8] [14].	Does not require a dedicated transmitter. Uses external illuminators of opportunity. Employs separate reference and surveillance channels; the received signal is only partially known [13] [18].
Signal Processing	Direct signal interference is minimal. Matched filtering with moderate complexity and straightforward synchronisation is used [14] [8].	Strong direct path interference must be suppressed. Uses cross-ambiguity function (CAF) based cross-correlation, which is computationally intensive [13] [34] [27].
Geometry & Coverage	Generally monostatic geometry with a single co-located transmitter–receiver pair [8].	Inherently bistatic or multistatic geometry, providing wider spatial coverage with multiple receivers [13] [18].
Detectability & Security	Uses a dedicated transmitter, making the system readily detectable and vulnerable to interception and electronic attack [9] [1].	Passive operation with no transmitter makes the system difficult to detect and inherently covert [9] [13].
Spectrum Usage	Requires a dedicated frequency allocation, contributing to spectrum congestion [1] [35].	Reuses existing broadcast or scientific transmissions and requires no dedicated spectrum allocation [13] [18] [24].
Interference & Electronic Warfare	Highly susceptible to jamming and electronic warfare; moderate clutter effects [8] [15].	More resilient to jamming due to passive operation, but suffers from strong clutter and multipath interference at the receiver [13] [19] [36].

1.7 Motivation for the Present Work

Active radar systems have been widely used for years in both civilian and military applications, but they have certain practical limitations like high power requirements, spectrum congestion, and vulnerability to electronic warfare techniques [15]. With development and technology the number of communication and broadcasting systems have increased rapidly. They can easily be used as IOOs in Passive Radar Systems [18]. Passive Radars have gained popularity especially due the advantages like covert operation, lower infrastructure cost, reduced power consumption, and efficient utilization of the electromagnetic spectrum [9], [24].

Another important reason for the growing interest in passive radar is the increasing requirement for distributed and multistatic sensing systems. In multistatic passive radar configurations, targets can be observed from multiple receiver locations, which improves spatial coverage and detection capability. Such configurations are particularly useful in situations where conventional monostatic radar systems may experience reduced performance [12], [20].

Recent progress in digital signal processing, software-defined radio systems, and computational hardware has also improved the practical feasibility of passive radar implementation [32], [33]. This study is particularly relevant from India's scientific and strategic requirement. India operates a unique set of high-power VHF scientific radar facilities that can be used as IOOs. The ARIES ST Radar in Nainital, Uttarakhand, operates at 206.5 MHz with a peak transmitted power of 235 kW [37]. The Advanced Indian MST Radar (AIR) at NARL Gadanki, Andhra Pradesh, operates at 53MHz with a total peak power of 1.024 MW, making it one of the most powerful VHF atmospheric radars in the world [38].

Moreover, India's rapidly growing space program has created an urgent national requirement for space situational awareness (SSA) capability. The population of trackable space debris in LEO currently exceeds 30,000 objects larger than 10 cm, with millions of smaller untracked objects posing increasing collision risk to operational satellites [21], [22].

Additionally, from a national security perspective, a passive radar network exploiting India's existing VHF infrastructure would provide covert wide-area surveillance capability with no additional spectrum requirements and a low probability of interception [9], [18].

India's dense urban environment also provides a rich source of FM, cellular, and broadcast IOOs that could support passive radar in border monitoring and air defence applications [9], [18].

The present work is motivated by the increasing importance of passive radar systems in modern surveillance applications and the need to study their practical and theoretical aspects in greater detail. The work focuses on simulation of passive radar systems using illuminators of opportunity, with emphasis on bistatic and multistatic radar geometry, and signal processing techniques used for target detection and localization.

1.8 Chapter Summary

This chapter introduces the fundamental concepts of radar systems and established the motivation for the present work. The term RADAR (Radio Detection and Ranging) was defined. The basic components of a radar system were reviewed, including the transmitter, duplexer, antenna, receiver, and signal processor, along with the physical quantities that can be measured by a radar system: target range, direction, radial velocity, and, in advanced systems, size and shape.

A concise history of radar development was presented. Followed by, the three fundamental radar configurations: monostatic radar (co-located transmitter and receiver), bistatic radar (geographically separated transmitter and receiver), and multistatic radar(multiple transmitter-receiver pairs) along with passive radars(radars that dontt have a dedicated trasnmission). The key advantages of bistatic and passive radar like covert operation, no spectrum allocation requirement, and ability to exploit existing infrastructure were discussed along with their limitations, including the bistatic position ambiguity and the complexity of direct signal interference cancellation.Advantages of multistatic radar were also discussed like, localization target of target and spatial coverage. Recent developments and in bistatic and monostatic radar along with operational bistatic and monostaic radars especially passive radars have also been mentioned.

A comparison of active and passive radar architectures was provided, covering differences in waveform control, signal processing complexity, system cost, and operational detectability. The chapter concluded with the motivation for the present work and identifying the ARIES ST Radar at Nainital(206.5 MHz, 235 kW) as a suitable illuminators of opportunity,

Chapter 2

Basic Radar Theory

In this chapter, we will provide an introduction to the sub-components of a radar system.

2.1 Basics of Antenna

2.1.1 Isotropic Radiator

An Isotropic Radiator is "a hypothetical lossless antenna having equal radiation in all directions" [16]. The power density radiated by an isotropic antenna with input power P_t(W) at a distance R(m) is [17]

$$S = \frac{P_t}{4\pi R^2} \tag{2.1}$$

Where, S(W/m^2) is the Power Density.

2.1.2 Gain

The ability of an antenna to direct the power into radiation in a particular direction is known as its gain. It is measured at the peak radiation intensity [17]. The radiation intensity is a far-field parameter. It is "the power radiated from an antenna per unit solid angle", in a given direction [16]. The Gain of a practical antenna increases its power density in the direction of its peak radiation [17]:

$$S = \frac{P_t G_t}{4\pi R^2} \tag{2.2}$$

Where S(W/m^2)is the Power Density of a Practical antenna that transmits power P(W) and has a gain G.

2.1.3 Radiation Pattern

A radiation pattern of an antenna is the representation of the radiation properties of the antenna as a function of space coordinates. It is also known as antenna pattern [16].

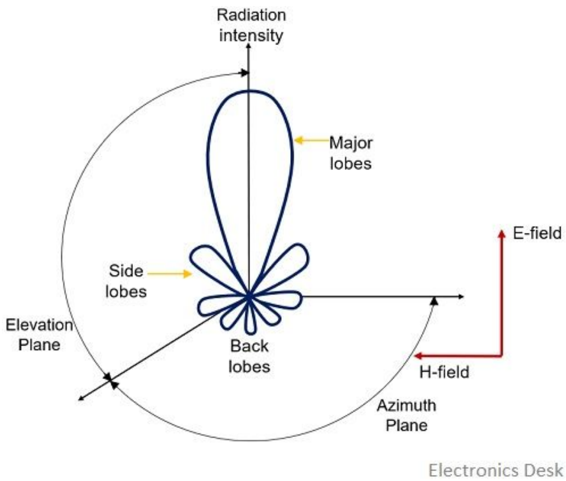

FIGURE 2.1: Figure 2.1 The radiation pattern of an antenna [5].

Key components of the radiation pattern include

Main Lobe - It is the region where the antenna radiates maximum power. It represents the desired direction of transmission and contains the peak radiation intensity.

Side Lobes - They are mainly produced because of the finite size of the antenna and interference effects between radiating elements. In antenna design, side lobes must be minimized to reduce unwanted radiation and interference.

Back Lobe - Radiation in the direction opposite to the main lobe.

E-plane - Plane containing the electric field vector and direction of propagation.

H-plane - Plane containing the magnetic field vector.

2.1.4 Directivity

The directivity of an antenna is defined as the ratio of the radiation intensity in a given direction from the antenna to the radiation intensity averaged over all directions [16].

$$D = \frac{U}{U_0} = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}} \tag{2.3}$$

$$D_{\text{max}} = D_0 = \frac{U|_{\text{max}}}{U_0} = \frac{U_{\text{max}}}{U_0} = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}} \tag{2.4}$$

Where, D = Directivity

D_0 = Maximum Directivity

U = Radiation Intensity

U_{max} = Maximum Radiation Intensity

U_0 = Radiation Intensity of Isotropic Source

P_{rad} = Total Radiated Power

2.1.5 Beamwidth

A Half Power Beamwidth (HPBW) is the angle between the two directions in which the radiation intensity is one-half the maximum value [16].

2.1.6 Bandwidth

The bandwidth of an antenna is defined as the range of frequencies within which the performance of the antenna, with respect to some characteristics like gain, return loss, etc., conforms to a specified standard [16].

2.1.7 Antenna Effective Aperture

Antennas capture power from the waves that are passing. The power that reaches the receiver depends on the Effective Area of the antenna [17]. The effective aperture A_e of an antenna is the area that would intercept the same power from an incident wave as the antenna actually receives. It is related to the antenna gain by [16]:

$$A_e = \frac{\lambda^2 G}{4\pi} \tag{2.5}$$

2.1.8 Radar Cross Section

The Radar Cross Section (RCS)(m^2) is a far-field parameter used to characterize the scattering properties of a radar target. It depends on the shape, material, and orientation of the object [8]. RCS depends not just on reflected EM waves from the target but also on how much of the wave is intercepted by the target and how much is reflected towards the receiver. So, interception, reflection and directivity together determine the RCS of a target. If the RCS of a target is made as low as possible it can be made "invisible" to the radar [10]. Similarly, to detect an object we need a good RCS, so the radar has to be sensitive.

2.2 The Radar System

2.2.1 Monostatic Radar Equation

As discussed in Chapter 1, in a monostatic radar system the transmitter and receiver are co-located and share a common antenna. Therefore, the range from the radar to the target R(m) is the same for both transmission and reception [8].

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 R^2} \tag{2.6}$$

Here, G_r is the gain of receiver.

2.2.2 Bistatic Radar Equation

According to eq(2.2), for a radar system, the target acts as a scattering object that intercepts and reradiates the transmitted electromagnetic energy. The transmitted power density at the target, at a distance of $R_t(m)$from the transmitter location is:

$$S_t = \frac{P_t G_t}{4\pi R_t^2} \tag{2.7}$$

If the target has RCS $\sigma(m^2)$, then the reradiated power or echo is

$$P_\sigma = S_t \sigma \tag{2.8}$$

The scattered power density at the receiver, which is at a distance R_r is

$$S_r = \frac{P_t G_t \sigma}{(4\pi)^2 R_t^2 R_r^2} \tag{2.9}$$

The received power by the receiver, with antenna effective aperture A_e is

$$P_r = S_r A_e \tag{2.10}$$

Substituting the value of A_e from eq(2.6),

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R_t^2 R_r^2} = \frac{P_{\text{EIRP}} G_t \lambda^2 \sigma}{(4\pi)^3 R_t^2 R_r^2} \tag{2.11}$$

2.2.3 Bistatic Range Sum

In a bistatic radar system, the transmitted signal travels from the transmitter to the target and then from the target to the receiver. Therefore, the total propagation distance is equal to the sum of the transmitter-to-target distance and the target-to-receiver distance. [18], [13] This total distance is called the bistatic range sum:

$$L = R_T + R_R \tag{2.12}$$

Where, R_T(m) = Transmitter-to-Target distance, R_R(m) = Target-to-Receiver distance

The propagation delay, τ(sec) is

$$\tau = \frac{R_T + R_R}{c} \tag{2.13}$$

Where, c = Speed of Light

All points that satisfy the property :

$$R_T + R_R = \text{constant} \tag{2.14}$$

form a geometric ellipse, where the transmitter and the receiver are at the focii. [18]

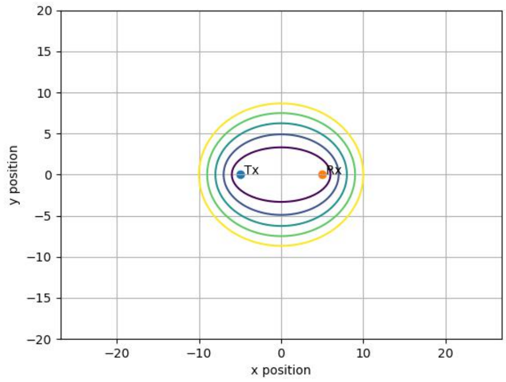

FIGURE 2.2: Family of contours with the transmitter and receivers and focii

The ellipses in Fig (2.2) are for the shown bistatic radar pair. Each contour represents the locus of all points with the same bistatic range sum $L = R_T + R_R$. The transmitter (T_x) and receiver (R_x) are the two focii of the ellipses. A single bistatic delay measurement constrains the target to one ellipse but cannot determine where on the ellipse the target is located, resulting in position ambiguity

2.2.4 Bistatic Doppler

In a bistatic radar system, the Doppler frequency shift is produced due to the motion of the target, which changes the total propagation path length between the transmitter, target, and receiver [18], [14]

$$f_d = \frac{1}{\lambda}\frac{d}{dt}(R_T + R_R) \tag{2.15}$$

which becomes

$$f_d = \frac{1}{\lambda}(\hat{R}_T + \hat{R}_R) \cdot \mathbf{v} \tag{2.16}$$

Where, R_T = Unit Vector from Target to Transmitter, R_R = Unit Vector from Target to Receiver, and v(m/sec) = Target Velocity

Since the total path is $R_T + R_R$, Doppler shift depends on how fast this sum changes with respect to time. [14] Unlike monostatic radar systems, where the Doppler frequency depends only on the radial component of target velocity, bistatic radar Doppler depends on both the transmitter-target geometry and the receiver-target geometry. [18]

If we are operating at a frequency $f = 206.5MHZ$ and we want to detect a Low Earth Orbit object at an altitude of around 1000 km and with orbital velocity $v = 7.5 \times 10 * 3m/s$

The wavelength corresponding to this frequency is calculated using

$$\lambda = \frac{c}{f} \tag{2.17}$$

Substituting the values,

$$\lambda = \frac{3 \times 10^8}{206.5 \times 10^6} \tag{2.18}$$

$$\lambda \approx 1.45 \text{ m} \tag{2.19}$$

For maximum Doppler shift, assume that the target motion is aligned with the bistatic observation geometry. In this case,

$$\vec{v} \parallel \left(\hat{R}_T + \hat{R}_R \right) \tag{2.20}$$

Therefore, the dot product becomes maximum:

$$\vec{v} \cdot \left(\hat{R}_T + \hat{R}_R\right) = v \left|\hat{R}_T + \hat{R}_R\right| \tag{2.21}$$

assuming $R_T \approx R_R$

$$f_D \approx \frac{2v}{\lambda} \tag{2.22}$$

Substituting the values,

$$f_D \approx \frac{2 \times 7.5 \times 10^3}{1.45} \tag{2.23}$$

$$f_D \approx 1.03 \times 10^4 \text{ Hz} \tag{2.24}$$

Therefore,

$$f_D \approx 10.3 \text{ kHz} \tag{2.25}$$

Thus, a LEO object at an altitude near 1000 km can produce Doppler shifts of the order of several kilohertz at 206.5 MHz. Such Doppler shifts are significantly larger than those produced by most terrestrial targets.

2.3 Noise and Detection

2.3.1 Thermal Noise (Johnson–Nyquist Noise)

Noise is unwanted electromagnetic energy that impairs a receiver's ability to detect the desired signal. It may originate within the receiver itself or may enter through the receiving antenna together with the desired signal [15]. Even in the absence of external interference, a receiver still produces a certain amount of unavoidable noise because of the random thermal motion of electrons in conductive materials. This type of noise is known as thermal noise or Johnson–Nyquist noise. [15]

Thermal noise power is given by

$$P_n = kTB \tag{2.26}$$

Where, P_n = Antenna Noise Power(W), k = Boltzman's Constant $(1.38\times10^{-23})(JK^{-1})$, T = Antenna Temperature (K), B = Bandwidth (Hz)

The equation shows that thermal noise power increases linearly with both temperature and bandwidth. Therefore, wider receiver bandwidths generally result in higher noise power. [15]

2.3.2 Signal-to-Noise Ratio

In a radar system, the received signal is always accompanied by noise.In a radar system, external noise is a strong function of the direction in which the radar antenna is pointed [15].

The ratio between the useful signal power and the noise power is called the Signal-to-Noise Ratio (SNR). SNR is the ability of a system to differentiate between signals and background noise [15]. SNR is given by

$$\mathrm{SNR} = \frac{P_s}{P_n} = \frac{P_{EIRP}\sigma_b G_r \lambda^2}{(4\pi)^3 k_B T_{sys} R_T^2 R_R^2} \tag{2.27}$$

A higher SNR indicates that the received signal is much stronger than the background noise, making signal detection easier. Similarly, low SNR values make it difficult to distinguish the desired signal from the surrounding noise.

2.3.3 Noise Figure

The Noise Figure describes the amount by which a receiver degrades the signal-to-noise ratio of an incoming signal [15]. In practical receiver systems, additional noise is introduced by electronic components such as amplifiers, mixers, and filters. Because of this, the output signal-to-noise ratio becomes smaller than the input signal-to-noise ratio. The Noise Figure is defined as:

$$F = \frac{(SNR)_{\text{in}}}{(SNR)_{\text{out}}} \tag{2.28}$$

Where, F = Noise Figure, $(SNR)_{\text{in}}$ = Input Signal-to-Noise Ratio, $(SNR)_{\text{out}}$ = Output signal-to-noise ratio

In decibels

$$\text{NF} = 10 \log_{10}(F) \quad \text{(dB)} \tag{2.29}$$

A noise figure of 0 dB ($F = 1$) would mean that the receiver does not add any noise at all and the output SNR is equal to the input SNR. This is the ideal lossless, noiseless receiver, which is practically not possible [15].

At FM broadcast frequencies, the external noise due to man-made interference and Galactic background typically dominates the thermal noise floor, reaching values of 10-40 dB above thermal noise [36].

Malanowski reports that the FM-based passive radar demonstrator, developed at Warsaw University of Technology, achieved a measured noise figure of 6 dB for the entire receiving chain [39]. Any value less than or equal to this can be treated as an achievable noise figure for a well-designed passive radar receiver.

Howland et al. report that in their FM bistatic passive radar implementation, the principal hardware limitation on system performance is the dynamic range of the analogue-to-digital converter, which limits the cancellation of the direct path interference that can be up to 90 dB greater than the target echo [19]

2.3.4 Friis Noise Formula

In practical receiver systems, multiple electronic stages are connected in cascade. The overall noise performance of such a system depends not only on the noise figure of each stage but also on the gain provided by the earlier stages [15]. as can be seen in Fig (2.3) the contribution of later stages to the total system noise becomes smaller because the signal is amplified as it passes through the preceding stages. As a result, the first few stages of the receiver play an important role in determining the overall noise performance.

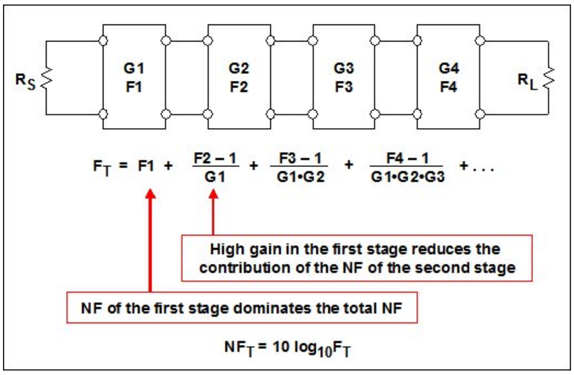

FIGURE 2.3: Cascade Noise figure [6]

2.3.5 Received Signal Power

A practical receiver introduces additional internal noise. This extra noise is represented using the equivalent noise temperature T_e(K). Hence, the total input-referred noise temperature becomes

$$T_{\text{total}} = T_0 + T_e \tag{2.30}$$

The output noise power becomes

$$N_{\text{out}} = Gk(T_0 + T_e)B \tag{2.31}$$

The output signal power is

$$S_{\text{out}} = GS_{\text{in}} \tag{2.32}$$

Input SNR:

$$SNR_{\text{in}} = \frac{S_{\text{in}}}{kT_0B} \tag{2.33}$$

Output SNR:

$$SNR_{\text{out}} = \frac{GS_{\text{in}}}{Gk(T_0 + T_e)B} = \frac{S_{\text{in}}}{k(T_0 + T_e)B} \tag{2.34}$$

Substituting into the definition of noise figure:

$$F = \frac{\frac{S_{\text{in}}}{kT_0B}}{\frac{S_{\text{in}}}{k(T_0+T_e)B}} \tag{2.35}$$

This gives us:

$$F = 1 + \frac{T_e}{T_0} \tag{2.36}$$

Rearranging

$$T_e = T_0(F - 1) \tag{2.37}$$

For a cascaded receiver system, the total equivalent input noise temperature can be expressed as [40].

$$T_{\text{sys}} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \cdots \tag{2.38}$$

Using eq(2.26) and substituting in eq(2.27)

$$T_{\text{sys}} = (F_1 - 1)T_0 + \frac{(F_2 - 1)T_0}{G_1} + \frac{(F_3 - 1)T_0}{G_1 G_2} + \cdots \tag{2.39}$$

The overall system noise figure is

$$F_{\text{sys}} = 1 + \frac{T_{sys}}{T_0} \tag{2.40}$$

Substituting in eq(2.28) and simplifying gives us

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots \tag{2.41}$$

2.3.6 System Noise Temperature

In a practical radar receiver, the total noise observed at the receiver input is contributed by several different sources [15]. These include noise received from the surrounding environment, losses in the transmission path, and noise generated internally by receiver components.

$$T_{\text{sys}} = T_{\text{ant}} + T_{\text{feed}} + T_{\text{rx}} \tag{2.42}$$

Where, T_{sys} = Total System Noise Temperature T_{ant} = antenna noise temperature T_{feed} = noise contribution due to feed line losses T_{rx} = receiver noise temperature

2.4 Antenna Array

The radiation pattern of a single element is relatively wide, and each element provides low values of directivity (gain). In many applications it is necessary to design antennas with very directive characteristics (very high gains) to meet the demands of long distance communication. This can only be accomplished by increasing the electrical size of the antenna. Enlarging the dimensions of single elements often leads to more directive characteristics. Another way to enlarge the dimensions of the antenna, without necessarily increasing the size of the individual elements, is to form an assembly of radiating elements in an electrical and geometrical configuration. This new antenna, is referred to as an array. Generally, the elements of an array are identical [17].

A single antenna radiates an electric field:

$$\mathbf{E} = E_\theta(\theta, \phi)\hat{\theta} + E_\phi(\theta, \phi)\hat{\phi} \tag{2.43}$$

where E_θ and E_ϕ are the two complex components (amplitude and phase) referred to some point on the antenna

Suppose that we have an array of antennas located at points r_1, r_2, and so on. We obtain the total pattern by adding the electric fields radiated from each:

$$E_{\text{total}} = \sum_{n=1}^{N} [E_{\theta i}(\theta,\phi)\hat{\theta} + E_{\phi i}(\theta,\phi)\hat{\phi}] e^{jk.r_i'} \tag{2.44}$$

2.4.1 Steering Vector

When a plane wave arrives at an angle , it reaches each antenna element at slightly different times because of the spatial separation (d) between adjacent elements. The extra path length between two adjacent elements is

$$\Delta L = d \sin\theta \tag{2.45}$$

Where, L = Path Difference, d = Distance Between the antennas, θ = Angle of Arrival

The corresponding phase difference is

$$\Delta\psi = \frac{2\pi}{\lambda} d \sin\theta \tag{2.46}$$

Where, $\Delta\psi$ = Phase Difference, $\frac{2\pi}{\lambda}$ = Wave Number

A steering vector represents the set of phase delays a plane wave experiences, evaluated at a set of array elements and normalized by root N

$$a(\theta) = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 \\ e^{i\psi} \\ e^{i2\psi} \\ \vdots \\ e^{i(N-1)kd\psi} \end{bmatrix} \tag{2.47}$$

Where, $a(\theta)$ = Steering Vector, N = Number of antennas in the array

2.4.2 Array Factor

The Array Factor is a function of the positions of the antennas in the array and the weights used [41]. By changing these parameters the antenna array's performance may be optimized to achieve desirable properties.

$$AF = \mathbf{w}^H \mathbf{a}(\theta) \tag{2.48}$$

Where, w = Weight Vector, H= Hermitian Transpose, $a(\theta)$ = Steering Vector

Array factor for XY plane is given by:

$$a(\theta, \phi) = \frac{1}{\sqrt{N}} \begin{bmatrix} e^{-jk(x_1 u_x + y_1 u_y)} \\ e^{-jk(x_2 u_x + y_2 u_y)} \\ \vdots \\ e^{-jk(x_N u_x + y_N u_y)} \end{bmatrix} \tag{2.49}$$

2.4.3 Grating Lobes

In an antenna array, the radiated waves from individual elements combine with one another. Under certain conditions, this combination can produce additional unwanted peaks in the radiation pattern known as grating lobes [16] These lobes appear when more than one direction satisfies the condition for constructive interference.

$$d \leq \frac{\lambda}{2} \tag{2.50}$$

This leads to the spacing constraint [17]

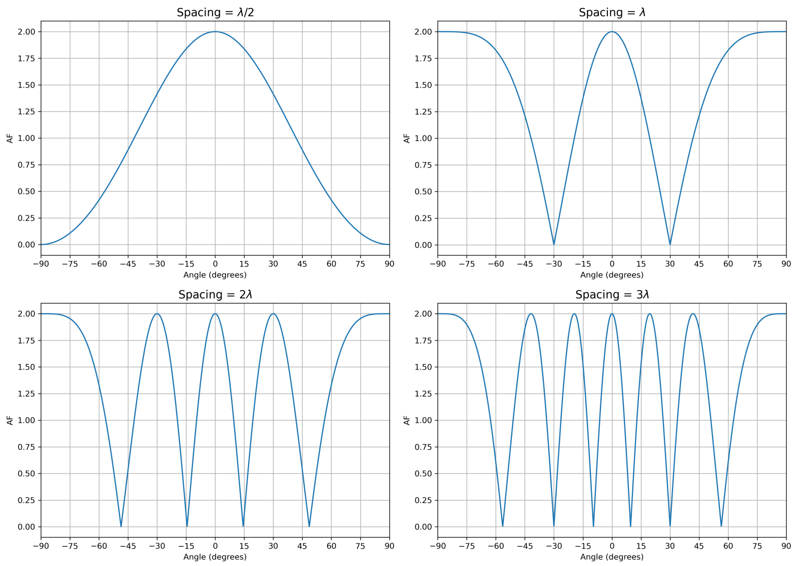

FIGURE 2.4: Plots for showing Grating Lobes in a phased array(i) Spacing is $\frac{\lambda}{2}$,(ii) Spacing is λ, (iii) Spacing is 2λ, (iv) Spacing is 3λ

Fig (2.4) shows how grating lobes start to occur in a 2 element array, as the spacing between them increases. If grating lobes are present, the antenna array receives signals with full gain from multiple directions simultaneously. This means that a signal arriving from the direction of a grating lobe produces the same output power as a signal arriving from the intended main beam direction. The array therefore cannot determine from which direction the signal actually originated, leading to direction ambiguity [16] [17].

2.4.4 Received Signal Power

The power intercepted by a single antenna element is given by eq (2.11)

For an array of antennas,

$$P_{\text{sig}} = NP_{sig,1} \tag{2.51}$$

So, the SNR becomes

$$\text{SNR} = \frac{P_s}{P_n} = \frac{P_{EIRP}\sigma_b G_r \lambda^2 N\tau}{(4\pi)^3 k_B T_{sys} R_T^2 R_R^2} \tag{2.52}$$

The receiver integrates the signal over a time interval. Therefore,

$$P_n = \frac{kTB}{\sqrt{B\tau}} \tag{2.53}$$

Where, τ = Time interval and $B\tau$ is the sampling rate

So, The SNR is now

$$\text{SNR} = \frac{P_s}{P_n} = \frac{P_{EIRP}\sigma_b G_r \lambda^2 N\sqrt{\tau}}{(4\pi)^3 k_B T_{sys} R_T^2 R_R^2 \sqrt{B}} \tag{2.54}$$

2.5 Chapter Summary

This chapter presented the basic terminologies required for understanding antenna and radar systems. The chapter began with the fundamental concepts of antenna theory, including isotropic radiators, gain, radiation pattern, directivity, beamwidth, bandwidth, and antenna effective aperture. Important antenna parameters such as radiation intensity and Radar Cross Section (RCS) were also discussed because of their importance in radar target detection and electromagnetic wave propagation.

The basic radar equations were then introduced. The monostatic radar equation and bistatic radar equation were derived to show how transmitted power, antenna gain, and RCS influence the received signal power. The concept of bistatic range sum and bistatic geometry was explained.

The concept of bistatic Doppler was introduced to explain how target motion produces frequency shifts in bistatic radar systems. Unlike monostatic radar, bistatic Doppler depends on both the transmitter-target and receiver-target geometries. An order of magnitude estimation was also performed for Low Earth Orbit (LEO) objects operating at 206.5 MHz, showing that Doppler shifts of the order of several kilohertz can be expected for high-velocity orbital targets.

The chapter further discussed receiver noise and radar detection concepts. Thermal noise, Signal-to-Noise Ratio (SNR), Noise Figure, Friis cascade noise formula, and system noise temperature were introduced to explain the limitations involved in weak-signal detection systems. The effect of receiver bandwidth, receiver temperature, cascaded receiver stages, and integration time on the achievable SNR was also discussed. Since passive radar systems rely on non-cooperative illuminators of opportunity and often operate with extremely weak reflected signals, it is important to understand receiver noise and system sensitivity.

The final part of the chapter focused on antenna arrays and phased-array systems. The limitations of single-element antennas were discussed, motivating the use of antenna arrays for improving gain and directivity. Steering vectors and array factors were introduced to describe how phase differences between antenna elements can be used for beamforming and beam steering. The chapter also showed that the received signal power increases with the number of antenna elements, which directly improves the Signal-to-Noise Ratio of the receiver system. These concepts form the theoretical basis for the phased-array simulations, passive radar analysis, and multistatic radar studies presented in the later chapters.

Chapter 3

Methodology

3.1 FM Signals

As discussed in earlier chapters, a passive radar system requires illuminators of opportunity. FM broadcasts are used as illuminators of opportunity in passive radar systems as they have continuous transmission and wide geographical coverage [18], [19] FM broadcast stations in Kanpur were chosen as transmitters for the system. The Frequency Transmission is in the range 88-108 MHz.

3.2 Radio Stations in Kanpur

Table 3.1: FM Radio Stations and Operating Frequencies

Radio Station	**Operating Frequency**
BIG FM	92.7 MHz
Radio Mirchi	98.3 MHz
Air Rainbow	102.2 MHz
Radio City	104.8 MHz
Red FM	93.5 MHz
Fever FM	95 MHz
Radio Gyan Vani	103.7 MHz

3.3 Yagi Uda Antenna

A Yagi-Uda antenna is a directional antenna consisting of 2 or more parallel resonant elements arranged along a common boom [16], [17]. It is easy to construct and has a gain typically higher than 10 dB. Yagi antennas generally operate in the HF to UHF band(3 MHz - 3 GHz).

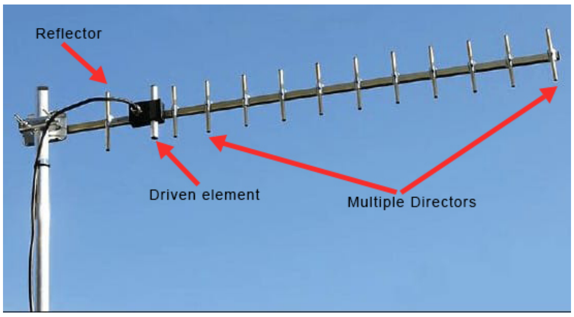

FIGURE 3.1: A 13-element Yagi Uda Antenna with 1 Reflector, 1 Driven element and 11 Driver Elements [7]

The Yagi-Uda antenna consists of multiple linear dipole elements arranged along a common boom, where one element is directly excited by the feed transmission line while the remaining elements act as parasitic radiators through electromagnetic coupling. The directly excited element functions as the driven element, while the parasitic elements operate as reflectors or directors depending on their position and dimensions [16]. The reflector is placed behind the driven element and is slightly longer in length, while the directors are positioned in front of the driven element and are slightly shorter. These parasitic elements help shape the radiation pattern and improve the directional characteristics of the antenna [16], [17].

Yagi is popular for television reception, but it is also used in a number of other domestic and commercial applications where an RF antenna with high gain and directivity is needed [16]. A Yagi-Uda antenna was used in the passive radar setup. In order to design a Yagi-uda antenna that is sensitive to FM broadcast frequencies 4NEC2 software was used to first find the ideal lengths for the reflector, driver and driven elements and then find the expected F/B Ratio, Gain and SWR of the antenna.

3.4 4NEC2 Software

4NEC2 is an electromagnetic simulation software based on the Numerical Electromagnetics Code (NEC) method and is widely used for antenna modelling and analysis. It is used for creating, viewing, optimising, and checking 2D and 3D style antenna geometry structures and generating, displaying and/or comparing near/far-field radiation patterns.

This software was used to design a Yagi-Uda antenna for the FM broadcast band. It helped find the optimized values of the Driver, Driven Element and the Reflector of a 3 element Yagi-Uda Antenna. It showed the antenna gain, Standing Wave Ratio(SWR) and Forward-Back Ratio(F/B) for these values. The optimizer present in 4NEC2 used iterative simulations to find the optimized dimensions of the antenna. The optimized geometry was exported as .nec file which has the dimensions of the 3 elements and the spacing between them.

The optimized .nec file generated using 4NEC2 was simulated to analyze the radiation characteristics and behaviour of the designed three-element Yagi-Uda antenna operating near 100 MHz A[1].

[1] The author would like to acknowledge Ms. Jayasandhya Meenakshinathan for the contribution in .nec file.

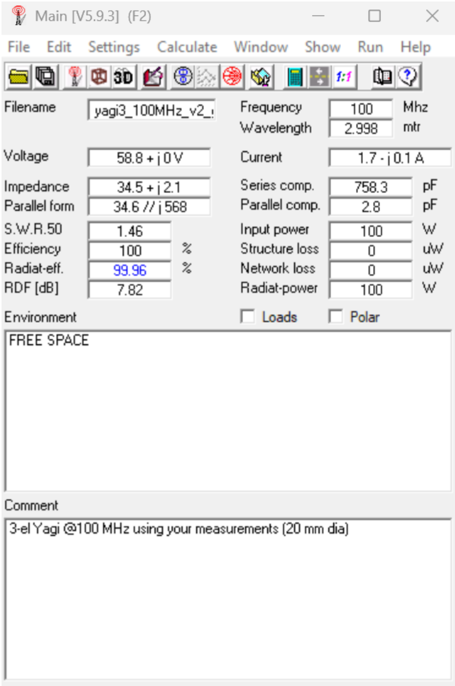

FIGURE 3.2: Screenshot of optimized .nec file for 100 MHz sensitive 3 element Yagi Uda Antenna

Fig (3.2) shows SWR = 1.46.

$$\text{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

[16] The reflection coefficient is calculated as

$$|\Gamma| = \frac{\text{SWR} - 1}{\text{SWR} + 1}$$

Substituting the value,

$$|\Gamma| = \frac{1.46 - 1}{1.46 + 1}$$

$$|\Gamma| = \frac{0.46}{2.46}$$

$$|\Gamma| \approx 0.187$$

The S_{11} in dB is given by

$$S_{11} = 20 \log_{10} |\Gamma|$$

$$S_{11} = 20 \log_{10}(0.187)$$

$$S_{11} \approx -14.56 \text{ dB}$$

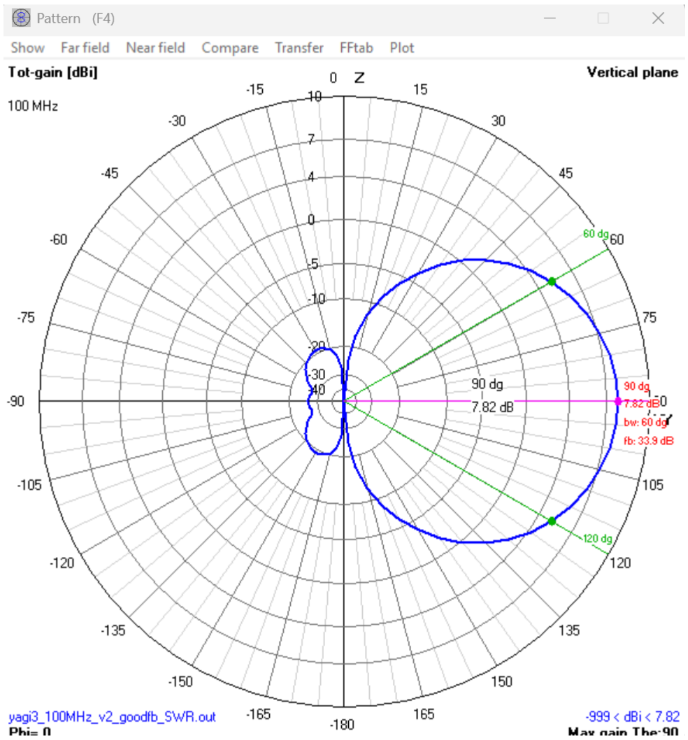

FIGURE 3.3: Simulated radiation pattern of the optimized three-element Yagi-Uda antenna generated using 4NEC2 software.

The simulated radiation pattern in Fig. 3.3 shows a well-defined main lobe directed toward the front of the antenna (in the direction of the directors) and a reduced radiation in the backward direction due to the reflector element. It can be observed from fig (3.4) that: Gain = 7.02 dB

Front-to-Back Ratio = 33.3 dB

Beamwidth = 60 deg

Front-to-back ratio is the ratio of radiated power in desired direction to the radiated power in it's opposite direction. It is the antennas ability to discriminate, against interfering signals coming directly from the rear, where the antenna is used for reception [42]. So, front-to-back (F/B) ratio of 33.3 dB indicates effective suppression of backward radiation by atleast 1000 times and good impedance matching to a 50 Ω feed. The half-power beamwidth (HPBW) of 60° is wide enough to survey a reasonable portion of the sky.These simulation results confirmed that the designed antenna was suitable for FM-band reception, and the geometry was used directly for fabrication.

3.5 Fabrication of 3-element Yagi-Uda Antenna

The optimized antenna dimensions obtained from the 4NEC2 simulations were used for the fabrication of a three-element Yagi-Uda antenna. The Antenna was fabricated using U-PVC pipes for the boom and support structure whereas, the measuring tape was used to make the Driver, Driven element and Reflector of the antenna. The driven element was cut from the middle and the 2 part were soldered to a coaxial cable. This was done as center feed has better balance and also results in lower standing wave ratio [42].

FIGURE 3.4: Fabricated three-element Yagi-Uda antenna

After fabrication, the antenna was experimentally tested by connecting it to a spectrum analyzer to evaluate its FM signal reception.

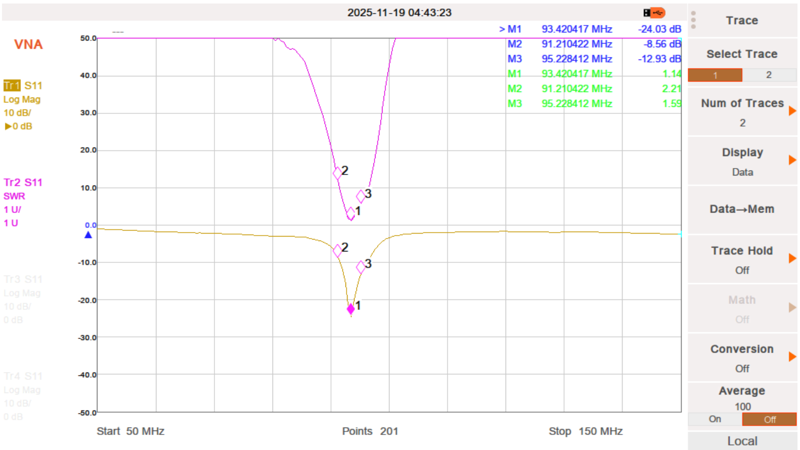

FIGURE 3.5: Fabricated three-element Yagi-Uda antenna connected to a Vector Network Analyzer (VNA)

As it can be seen in Fig 3.5 the reflection coefficient (S11) and Standing Wave Ratio (SWR) of the fabricated antenna were measured using a Vector Network Analyzer over the FM broadcast frequency range. The measured response shows a clear resonance near the FM broadcast band around 93–95 MHz, indicating that the antenna was successfully tuned for FM signal reception.The fabricated three-element Yagi-Uda antenna was experimentally characterized using a Vector Network Analyzer (VNA). The S11 parameter was used to determine the resonance characteristics of the antenna. The processed results showed a minimum reflection coefficient close to what was observed on the VNA. The measured S11 value is better than -25 dB.

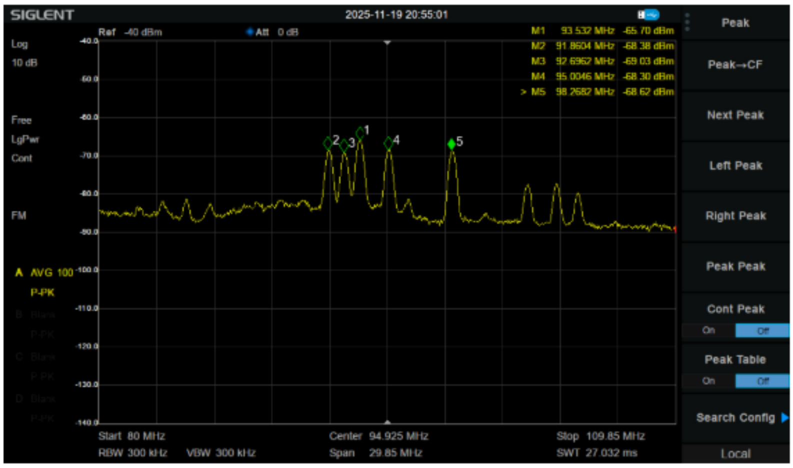

FIGURE 3.6: Fabricated three-element Yagi-Uda antenna connected to a spectrum analyzer for FM signal reception measurements.

As it can be observed in Fig 3.6 multiple FM broadcast carriers were successfully observed within the operating frequency range. Prominent peaks were detected near 91.86 MHz, 92.70 MHz, 93.53 MHz, 95.00 MHz, 98.27 MHz The received signal levels were approximately in the range of -65 dBm to -69 dBm. These frequencies correspond closely to locally available FM broadcast stations in Kanpur mentioned in Table 3.1. The measurements confirmed that the fabricated antenna was capable of receiving strong FM broadcast transmissions.

3.6 GNU Radio and Software Defined Radio

GNU Radio is an open-source software development framework widely used for Software Defined Radio (SDR) applications. GNU Radio consists of signal processing blocks that can be connected together using graphical flowgraphs or Python scripts. These blocks perform operations such as signal reception, filtering, modulation and demodulation, spectrum analysis, frequency translation, data recording, etc. In the current work, GNU Radio was used together with RTL-SDR signal acquisition and processing. First the RTL-SDR was connected to a signal generator, to give it a known source and validate the working of the setup.

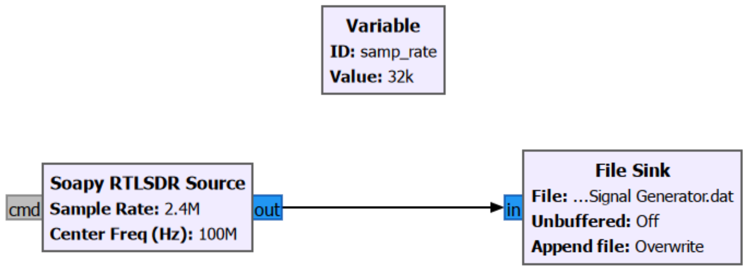

FIGURE 3.7: Blocks used in GNU Radio to save the data from the Yagi Uda Antenna through RTL-SDR

The "Soapy RTLSDR Source" is used to take input from RTL-SDR. The sample rate is 2400000 samples/sec and the centre frequency is 100MHz. The File Sink block is used to save the input data in .dat format.

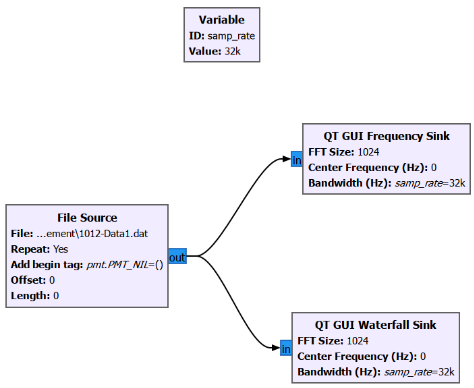

FIGURE 3.8: Blocks used in GNU Radio to observe the saved data

The “File source” block is used to process the previously recorded data. It is connected to “QT GUI Frequency Sink” and “QT GUI Waterfall Sink”, with FFT Size set to 1024. The following Graph was obtained from the two blocks.

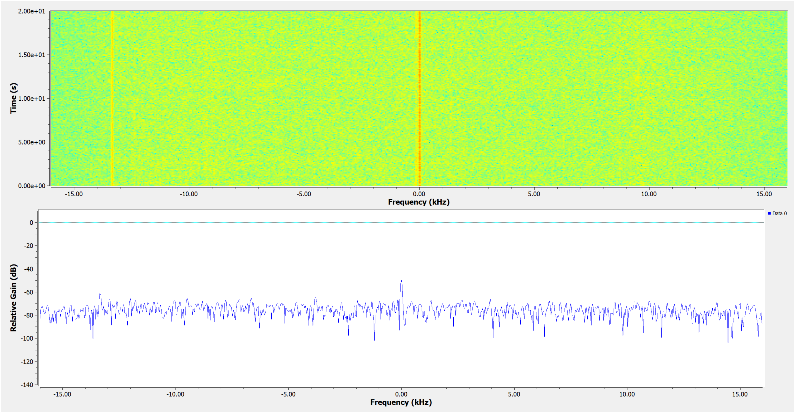

FIGURE 3.9: Time vs Frequency Graph and Gain vs Frequency Graph obtained from Fig (3.8)

The frequency-domain plot shows the distribution of signal power with frequency, while the waterfall plot displays the variation of signal intensity over time. The center frequency of the receiver was tunes to 100 MHz, and a strong narrowband peak can be observed near 0 kHz offset frequency. This corresponds to the received carrier signal after digital downconversion to baseband. The waterfall plot shows that the signal remains stable with time, appearing as a continuous vertical line throughout the observation interval.The surrounding region mainly consists of background receiver noise and low-level spectral fluctuations. A weaker secondary line is visible near -13 kHz offset, it could be due to local oscillator imperfections or residual interference present in the SDR receiver chain.

3.6.1 Limitations of FM-Based Passive Radar

The initial data acquisition was carried out using FM broadcast signals as illuminators of opportunity. Although strong direct-path FM signals were successfully received using the fabricated Yagi-Uda antenna, the FM was very strong and it was saturating the SDR. As a result, the study gradually expanded towards investigating alternative radar and observation facilities available in India that could support passive radar and multistatic radar related studies.

3.7 Using Existing Radar Facilities as Illuminators of Opportunity

During the later stages of the work, attention was directed toward understanding larger radio observation and radar-related facilities available in India that could potentially support passive sensing. Among these, the Aryabhatta Research Institute of Observational Sciences (ARIES), located in Nainital, Uttarakhand, was studied because of its proximity to Kanpur and high transmitting power. The Aryabhatta Research Institute of Observational Sciences (ARIES), located in Nainital, Uttarakhand, India, is a major research institute involved in astronomy, astrophysics, atmospheric sciences, and space science research. The institute operates several observational facilities including optical telescopes, atmospheric instruments, and radar systems used for scientific investigations [37]

TABLE 3.2: Specifications of the ARIES Radar System

Parameters	**Description**
Radar Location	29.4° N, 79.5° E, Magnetic Dip: 46.5°, 1793 m above mean sea level (amsl)
Operating Frequency	206.5 MHz with 5 MHz bandwidth
Total Peak Transmit Power	235 kW
Antenna Array	Near-circular 588 element Yagi–Uda antenna array with equilateral triangular grid
Antenna Array Aperture	490 m^2
Antenna Gain	34 dBi
One-way 3 dB Beam Width	3.3°
First Side-lobe Level (FSLL)	-17.7 dB with uniform distribution
Beam Scanning Capability	Azimuth: 0°–360° in steps of 1°; Off-Zenith: 0°–30° in steps of 1°
Peak Power of each TR Module	400 W (typical)
Maximum Duty Cycle	13%
Power Aperture Product	$\sim 1 \times 10^8$ W m^2
Pulse Width	Un-coded: 0.5–64 μs Coded: 0.5–64 μs with baud length varying from 0.25 μs to 64 μs
Highest Resolution	37.5 m
Pulse Repetition Frequency	250 Hz to 8 kHz
Receiver and Signal Processing	6-channel DDC-based system
Dynamic Range	70 dB
Maximum Number of Range Bins	512
Maximum FFT Points	2048
Modes of Operation	Doppler Beam Swinging (DBS), Spaced Antenna (SA)
Products and Data Storage	Wind Speed, Wind Direction, Moments (m_0, m_1, and m_2), U, V, W

3.8 Simulation Studies of the System

As mentioned in eq (2.54) SNR is directly proportional to EIRP, Radar Cross Section, Gain of Receiver, Wavelength, Number of Antennas and Sample Time. SNR is inversely proportional to System Temperature, Distance between Target and transmitter and Distance between Target and Receiver. Of all these parameters we can control the Number of antennas in our array and sample time. We can also control the Gain of the receiver array upto some extent, by making changes in it's dimensions. Yagi Uda Antenna is used in this study to maximize gain for the received signal. Below are some graphs which show how the SNR and Number of antennas vary with various parameters

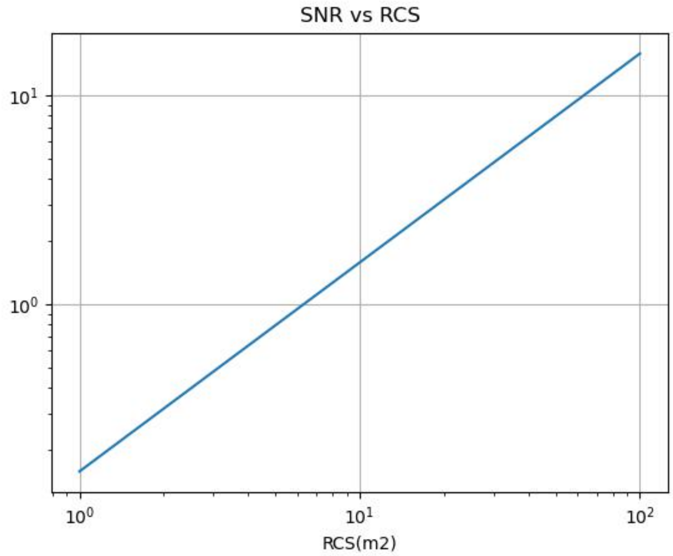

Figure 3.10: SNR(dB) as a function of target radar cross section (RCS)

Fig (3.10) shows the variation of SNR with RCS for where the no. of antennas are 10, sampling time is 0.5 seconds, bandwidth is 20 kHz and the $R_t = R_r = 3000km$. As given in eq (2.54) SNR is $\propto$ RCS, the plot gives a straight line with unit slope on the log-log axis.

The SNR increases significantly, with increase in Radar Cross-section. This means it is easier to detect targets with higher RCS like aircrafts and similarly it is difficult to detect targets with lower RCS like small debris or drones.

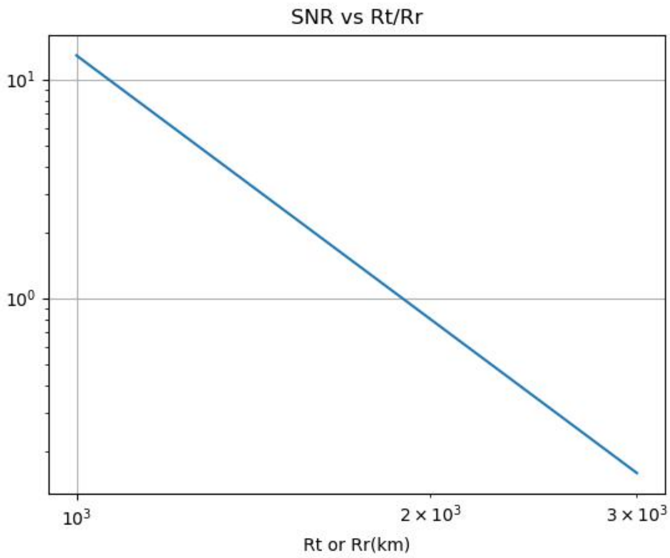

FIGURE 3.11: SNR as a function R_t or R_r

Fig (3.11) shows the SNR vs R_T or R_R plot. From eq(2.27) $SNR \propto \frac{1}{R_t^2}$ For 10 antennas, 0.5 seconds of sampling time, 20 kHz bandwidth and 1 m^2 RCS the plot has a downward steep slope. The SNR just reaches 10 dB when $R_t = R_r = 1000km$. Even a modest increase in the distance can cause severe SNR degradation. So, if the $R_T or R_r$ is quite high, it is important to increase either the number of antennas as well as their gain or sampling time, as these are the only factors that can be controlled while design a passive radar.

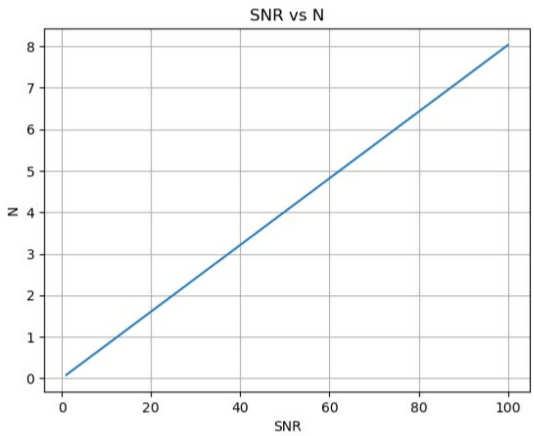

FIGURE 3.12: SNR as a function of the number antennas in a phased array

Fig. (3.12) shows how SNR increases with the number of antennas.As mentioned in eq(2.52) for a phased array, the N element signals add coherently and gives total power of $N * P_{single}$ hence being proprotional to SNR value. This plot is for 1 m^2 RCS, 0.5 seconds of sampling time, $R_t = R_r = 3000$ km and 20 kHz bandwidth. The plot shows that adding more antennas to the phased array will increase the SNR.

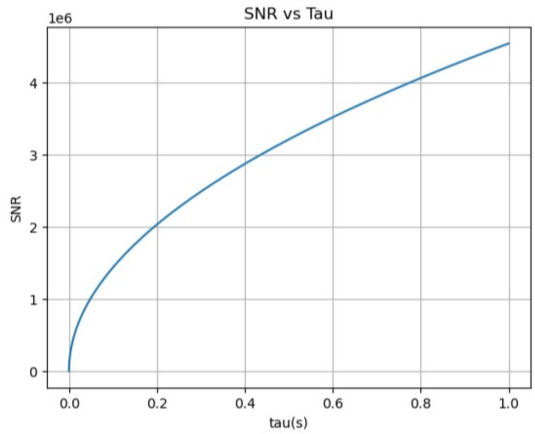

FIGURE 3.13: SNR as a function of integration/sample time τ

Fig. (3.13) shows the variation of SNR with integration time τ when N=10, $\sigma = 1$ m^2, $R_t = R_r = 3000$ km and is B = 20 kHz. Increasing the coherent integration time improves SNR, as according to the eq (2.54) $SNR \propto \sqrt{\tau}$.

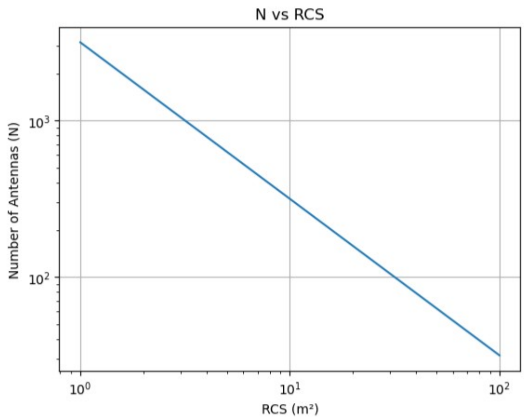

FIGURE 3.14: Minimum detectable RCS as a function of the number of receive array elements N

Fig. 3.14 shows how the RCS value decreases with increasing N when $\tau = 0.5$ s, R_t $= R_r = 3000$ km. the minimum detectable RCS can be found using the the eq(2.27)

$$\sigma_{b,\min} = \frac{(4\pi)^3 k_B T_{\text{sys}}\, \text{SNR}_{\min}\, R_T^2 R_R^2 \sqrt{B}}{P_{\text{EIRP}} G_r \lambda^2 N \sqrt{\tau}} \tag{3.1}$$

We want the minimum detectable RCS to be as low as possible as that would mean that the radar is able to detect the smallest of RCS.

These results clearly demonstrate that increasing the number of receive array elements significantly improves the sensitivity and detection capability of the passive radar system. From the plots it can be inferred that phased arrays improves the received signal power and thus the overall SNR of the radar system. Since passive radars receive weak reflected signals, the parameters that can be controlled like number of antennas, gain of the antennas(antenna with high directivity and gain are chosen for passive radars) and the sampling time. To have a high gain, Yagi-Uda antenna has already been chosen, but to further improve the overall performance of the system simulations were done, gradually increasing the number of antennas and looking at their overall gain and beam pattern.

3.9 Simulation of Phased Arrays

3.9.1 Simulation of Beam Pattern

As discussed in the previous section, different geometries and , number of antennas were observed. The simulations were done considering the ARIES radar frequency, 206.5 MHz. This gives the $\lambda = 1.452m$. According to the eq (2.50) the distance between the elements is $\lambda/2$ for all the arrays.

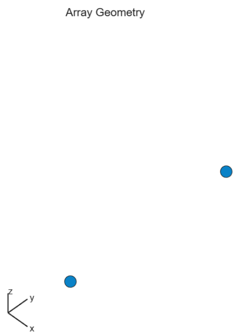

FIGURE 3.15: Phased Array with 2 elements - Array Geometry

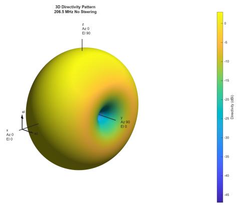

FIGURE 3.16: Phased Array with 2 elements - 3D Plot

Simulation with 2 antennas was done first. Fig (3.15 shows the linear geometry of the phased array. Fig (3.16) show the 3D plot of the beam pattern of the antenna, respectively. It is visible the array has very low directivity.

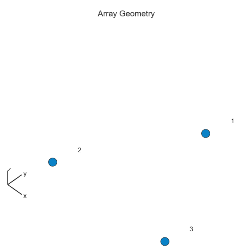

FIGURE 3.17: Phased Array with 2 elements - Array Geometry

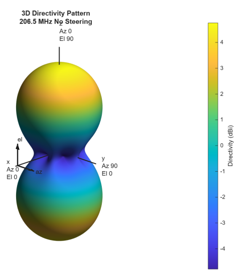

FIGURE 3.18: Phased Array with 3 elements - 3D Plot

To see if increasing the antennas will increase the directivity, simulation with 3 antennas was done. Fig (3.17) shows the triangular geometry of the phased array. Fig (3.18) show the 3D plot of the beam pattern of the antenna, respectively. The directivity has improved in comparison to the previous simulation.

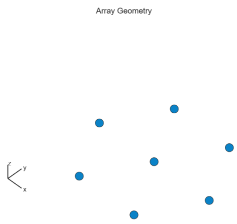

FIGURE 3.19: Phased Array with 7 elements - Array Geometry

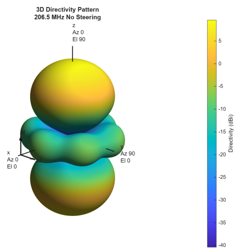

FIGURE 3.20: Phased Array with 7 elements - 3D Plot

To see if increasing the antennas will increase the directivity, simulation with 7 antennas was done. Fig (3.19) shows the hexagonal geometry of the phased array. Fig (3.20) show the 3D plot of the beam pattern of the antenna, respectively. The directivity has further improved in comparison to the previous array. The directivity of the array kept increasing as the number of antennas increased. These simulations were done without steering the the array in any direction. For practical experiments, the array will have to be steered in a particular direction. So, to see the beam pattern of a steered array, further simulations were done to see the beam pattern of array shown in Fig (3.19), which is a hexagon eometry. For simulation purpose azimuth was taken to be 30deg and elevation was taken to be 40deg

Fig (3.20) shows the steered beam pattern of the same array shown in Fig (3.19). The back lobe has significantly reduced and the main lobe is now towards the steered

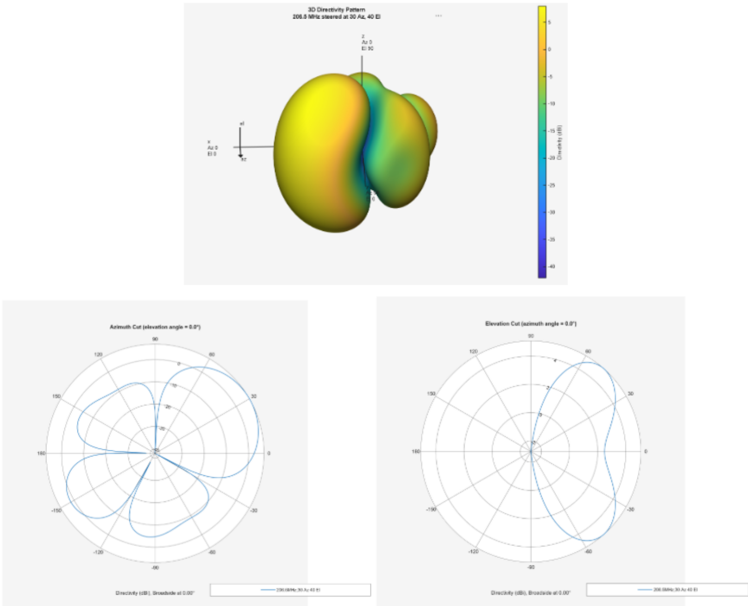

FIGURE 3.21: 7 Element antenna array(i) 3-d Steered Beam Pattern, (ii)2-d Steered Beam Pattern(Azimuth), (iii) 3-d Steered Beam Pattern(Elevation)

direction. Beam steering increases the efficiency of the array and is therefore essential for Passive Radar Systems.

3.10 Simulation of 7 element Yagi Uda Antenna

A 7-element Yagi has a better gain and directivity, in comaprison to a 3 element Yagi. The gain increases as more director elements are added to the Yagi. [42] [16]. Simulation of a 7 element Yagi was done, to find the relative increase in gain and directivity. The 4NEC2 software was used again to design a 7 element Yagi-Uda antenna for the 206.5 MHz frequency. It helped find the optimized values of the 5 Driver Elements, 1 Driven Element and 1 Reflector Element. The optimizer was configured to maximise gain and F/B ratio while maintaining SWR $<$ 1.5 at the operating frequency. It showed the antenna gain, Standing Wave Ratio(SWR) and Forward-Back Ratio(F/B) for these values.

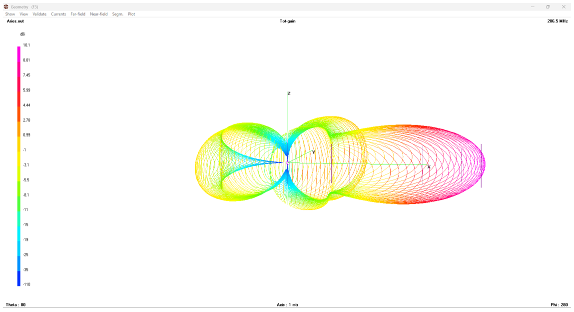

FIGURE 3.22: Beam Pattern of Optimized antenna

As seen in fig (3.22) the beam pattern shows strong directional radiation with a concentrated main lobe in the forward direction and a reduced backward radiation due to the reflector element. The simulated radiation pattern was also analyzed in both horizontal and vertical planes to study the directional characteristics and beam width of the antenna.

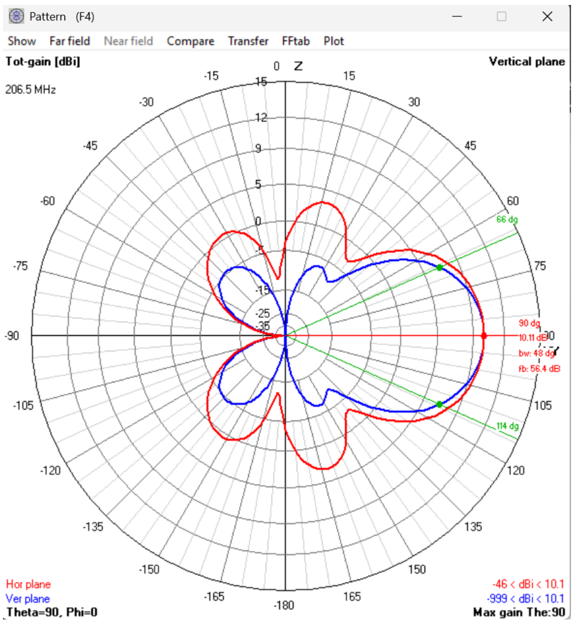

FIGURE 3.23: Radiation Pattern of optimized antenna in Horizontal and Vertical Plane

According to the Simulation result in Fig 3.23 we obtained

Gain = 10.11 dB

F/B = 56.39 dB

Beamwidth = 48 deg

SWR = 1.3906

It can be clearly seen that the 7 element Yagi Uda has a higher gain compared to the earlier 3-element Yagi-Uda antenna . It also has a higher F/B ratio in comparison with 3-element Yagi. Since, the simulation results were satisfactory the antenna was fabricated.

3.11 Fabrication of 7-element Yagi-Uda Antenna

The 7-element Yagi-Uda antenna was fabricated using the same construction techniques used for the earlier 3-Yagi.

Table 3.3: Dimensions of the 7-Element Yagi–Uda Antenna

Parameter	**Value**
Reflector Length	78.20 cm
Driven Element Length	63.86 cm
Director 1 Length	54.72 cm
Director 2 Length	54.27 cm
Director 3 Length	56.09 cm
Director 4 Length	47.12 cm
Director 5 Length	61.01 cm
Spacing between Reflector and Driven Element	47.43 cm
Spacing between Driven Element and Director 1	31.37 cm
Spacing between Director 1 and Director 2	13.14 cm
Spacing between Director 2 and Director 3	52.03 cm
Spacing between Director 3 and Director 4	27.76 cm
Spacing between Director 4 and Director 5	13.81 cm

The antenna was deployed on the terrace and the data was acquired continuously for 22 hrs.

Figure 3.24: Setup of 7-element Yagi Uda Antenna for Data Acquisition

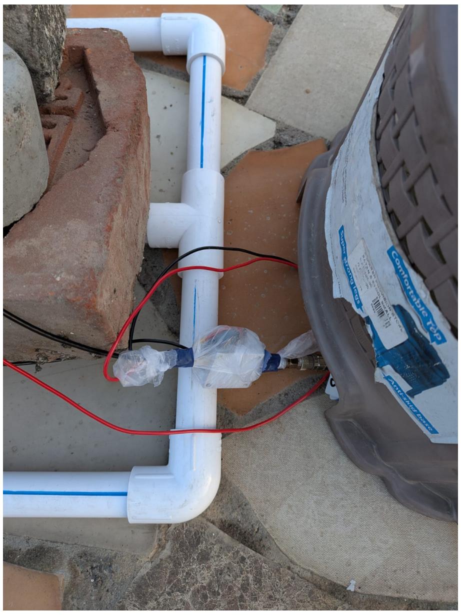

Figure 3.25: Covering the Bias Tee to protect it against the rain

FIGURE 3.26: Covering the LNA to protect it against the rain

The system was connected as follows.

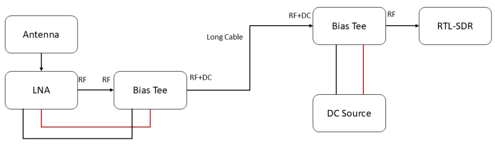

FIGURE 3.27: Block Diagram of connection of 7-element Yagi Uda Antenna to RTL-SDR

The coaxial cable connected to the driven element was kept as short as possible to reduce attenuation and was connected to a LNA to minimize noise. The received RF signals from the antenna were first amplified using the LNA in order to improve weak signal reception. A very short wire connected the antenna to the LNA, to reduce attenuation of the signal. The Bias Tee allowed the LNA to be powered through the same coaxial cable carrying the RF signal, avoiding the need for a separate power supply cable at the antenna.

The RTL-SDR receiver digitized the incoming RF signals and transferred the data to a laptop for recording and signal processing using GNU Radio. Data was acquired continuously for 22 hours, centered at 206.5 MHz with a sample rate of 1.024 MSPS and FFT size of 1024.

3.12 Chapter Summary

This chapter presented the methodology adopted for the development and analysis of a passive radar receiver system using FM broadcast signals and phased array concepts. Initially, FM radio stations operating in the 88-108 MHz band in Kanpur were identified as illuminators of opportunity for passive radar experiments. A three-element Yagi–Uda antenna was designed using 4NEC2 electromagnetic simulation software and optimized for operation near 100 MHz. The simulated antenna characteristics, including gain, beamwidth, front-to-back ratio, and SWR, demonstrated suitable performance for FM signal reception.

The optimized antenna was fabricated and experimentally characterized using a Vector Network Analyzer and spectrum analyzer. The measurements confirmed good impedance matching and successful reception of multiple FM broadcast stations. GNU Radio and RTL-SDR were then used for signal acquisition, visualization, and storage of received RF data.

The chapter further discussed the limitations encountered while using strong FM broadcast signals as illuminators, particularly SDR saturation due to strong direct-path interference. So, the attention was directed towards understanding larger radio observation and radar facilities available in India that could potentially support passive radar system. Its phased-array Yagi antenna system and radar specifications were reviewed as a potential high-power illuminator of opportunity.

Simulation studies based on the bistatic radar equation were then carried out to analyze the influence of radar cross section, range, integration time, and number of antennas on the signal-to-noise ratio. The results showed that increasing the number of array elements and coherent integration time significantly improves radar sensitivity and detection capability.

Further simulations were performed to study phased-array beam patterns using two-element, three-element, and seven-element antenna geometries. Beam steering simulations demonstrated improved directivity and suppression of back lobes. Finally, a seven-element Yagi–Uda antenna was designed, simulated, fabricated, and integrated with an LNA, bias tee, RTL-SDR, and GNU Radio based receiver setup for long-duration data acquisition at 206.5 MHz. The overall methodology established the feasibility of using phased array Yagi antennas for passive radar and radio observation studies.

Chapter 4

Observations and Results

4.1 Bistatic Passive Radar Simulations

Numerical simulations were performed in Python to characterise the bistatic radar system performance.

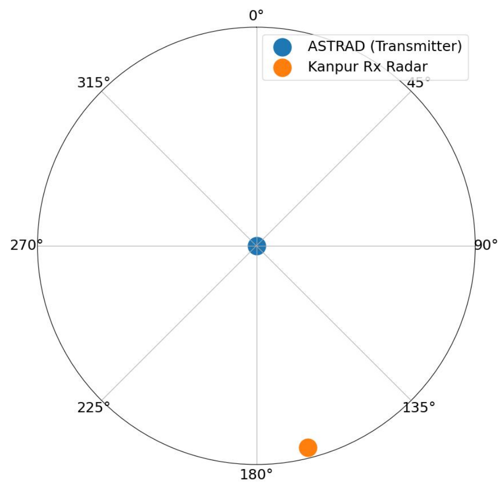

FIGURE 4.1: Bistatic geometry of the ARIES-Kanpur passive radar system

Fig (4.1) shows a polar plot with ARIES transmitter in the center, blue color. The plot shows the directions of the receiver, IIT Kanpur, orange color with respect to ARIES.

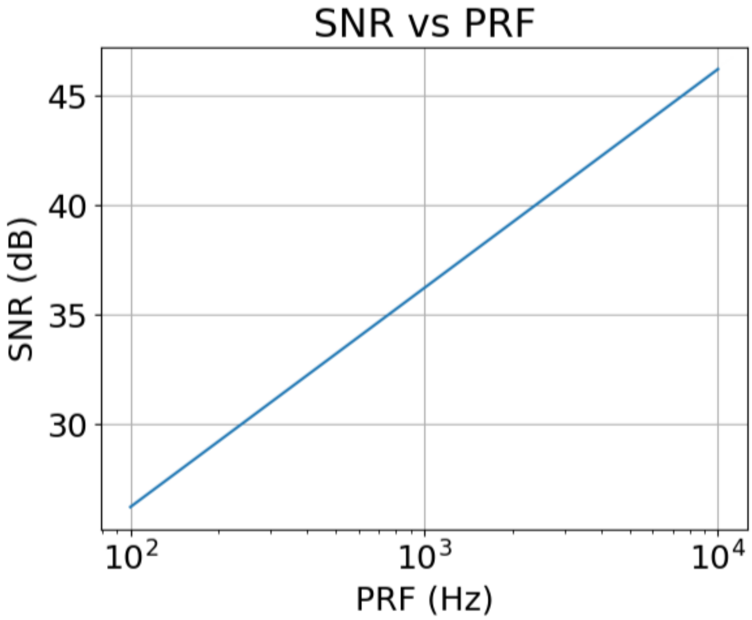

FIGURE 4.2: Bistatic SNR as a function of pulse repetition frequency (PRF) for the ARIES–Kanpur configuration.

The Pulse Repetition Frequency (PRF) is the number of radar pulses transmitted per second, measured in Hz. A higher PRF means more pulses illuminate the target per unit time. The average transmitted power equals peak power multiplied by pulse width multiplied by PRF, so SNR scales linearly with PRF [8] [14]. Fig (4.2) shows the straight line on the plot that confirms that SNR increases with PRF. This plot is significant because it shows how much the detection capability depends on the operating mode of ARIES.

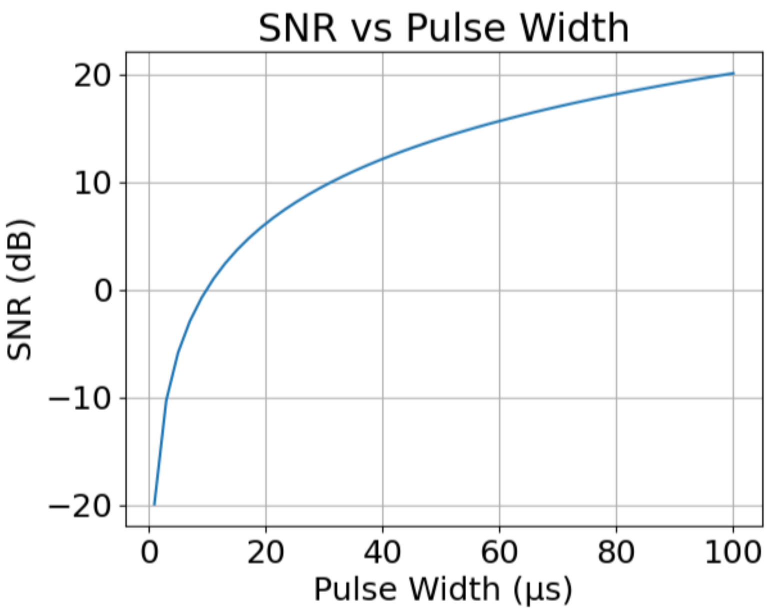

FIGURE 4.3: Bistatic SNR as a function of pulse width τ for the ARIES–Kanpur configuration.

Fig. 4.3 shows the variation of SNR with pulse width τ. The pulse width τ is the duration of each transmitted radar pulse in microseconds. Increasing the pulse width at fixed peak power increases the energy per pulse. Output SNR is proportional to pulse energy, so SNR increases with τ [14], [15].

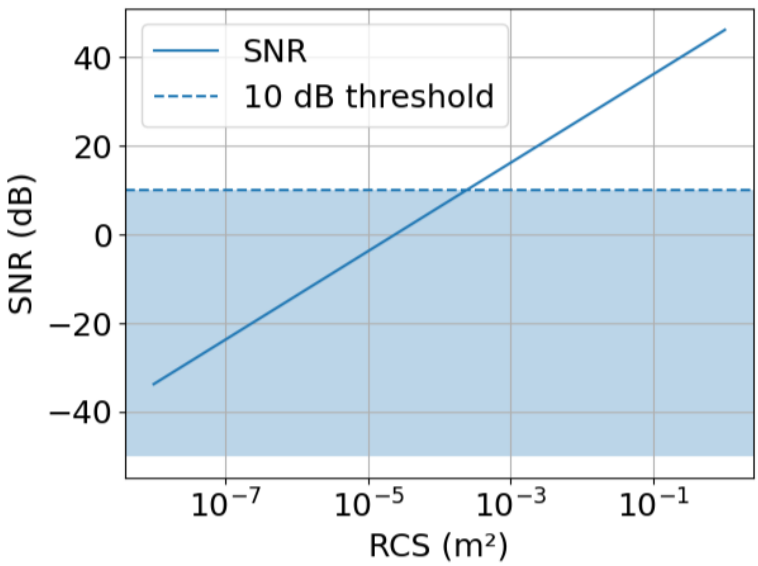

FIGURE 4.4: SNR vs target RCS for the bistatic ARIES–Kanpur system

The radar cross section (RCS) measured in m^2 describes how strongly a target reflects the radar signal. Larger, metallic objects have higher RCS; small debris objects have very low RCS. The Fig (4.4) shows a plot of SNR vs RCS. Since $SNR \propto \sigma$ the relationship is a straight line with unit slope on log-log axes. The 10 dB detection threshold is marked as a dashed horizontal line. 10dB is considered a standard threshold value [10], [43]. Below this threshold is the shaded non-detection zone, where the targets cannot be reliably detected. The threshold crossing point defines the minimum detectable RCS of the system under the simulated conditions.

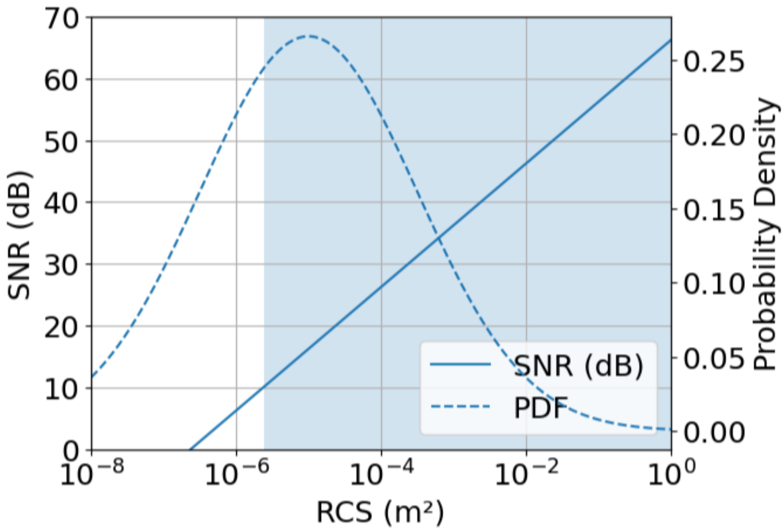

FIGURE 4.5: Bistatic SNR vs target RCS with the debris RCS probability density function (PDF) overlaid

The Fig (4.5) combines the SNR curve from Fig (4.4) with the probability density function of the expected space debris RCS distribution overlaid on the second y-axis. The PDF shows that most (centimeter-scale) debris objects are present in the RCS range of 10^{-6} to 10^{-4}.

The plot directly shows what fraction of the real debris population the system can detect. Objects to the right of the threshold crossing are in a detectable range, whereas those to the left are not.

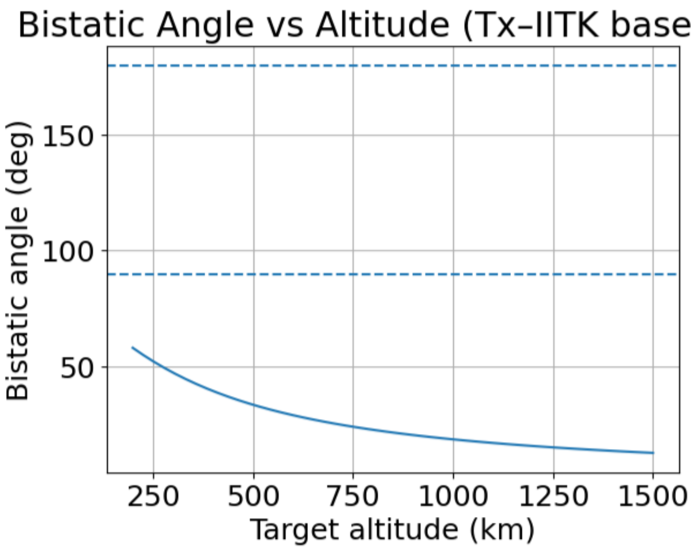

FIGURE 4.6: Bistatic angle as a function of target altitude for the ARIES–Kanpur bistatic pair.

The bistatic angle β is the angle at the target between the direction to the transmitter and the direction to the receiver [13]. For the ARIES-Kanpur geometry, β decreases from about 60 deg at 200 km altitude to about 15 deg at 1500 km. The two dashed reference lines at 90° and 180° mark the right-angle geometry and the forward scatter limit respectively. β approaching 180° would give forward scatter RCS enhancement [1], but the ARIES-Kanpur geometry never reaches this.To achieve β=180°, the receiver must be placed such that ARIES, the target, and the receiver are in a straight line, with the target in the middle.
For a target with speed $= v$, λ is the wavelength, and δ is the angle between the target velocity vector and the bistatic bisector, The Doppler shift is given by [13] [15]:

$$f_d = \frac{2v}{\lambda} \cos\left(\frac{\beta}{2}\right) \cos\delta \tag{4.1}$$

At low altitudes ($\beta \approx 60$ deg), the system operates in a geometry where bistatic RCS sensitivity is enhanced, while at high altitudes ($\beta \approx 15$ deg), $\cos(\beta/2) \approx 0.99$ and the Doppler sensitivity approaches its monostatic maximum value [13], [15].

Since these simulation results were satisfactory, it was decided to move forward with a simulation and then the fabrication for 7-element Yagi antenna, as mentioned in the previous chapter.

4.2 Data Acquisition through RTL-SDR and GNU Radio

The data was acquired for 22 hours through the 7-element Yagi-Uda antenna. The RTL-SDR was connected to the antenna via a Low Noise Amplifier (LNA) and Bias Tee. The center frequency was set to 206.5 MHz. Data was acquired and processed in real time using the GNU Radio flowgraph shown in Fig (4.1).[1]

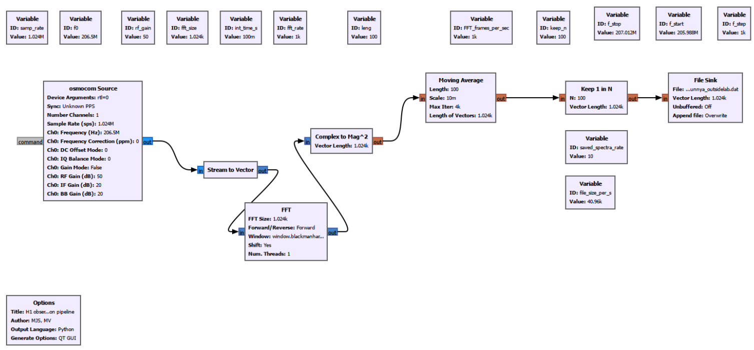

FIGURE 4.7: GNU Radio flowgraph used for the 22-hour data acquisition at 206.5 MHz.

The Sample Rate was set to $1.024e6$ that means $1.024 * 10^6 samples/s$ with 206.5 MHz as the center frequency. The FFT Size was taken to be 1024.

The "Osmocon Source" block acts as an interface between RTL-SDR and GNU Radio. It outputs complex baseband samples and feeds it to "Stream to Vector" block which then converts the continuous sample stream into vectors. Each vector becomes a FFT frame. In this case fixed-length vectors of 1024 samples. The output is then passed to the "FFT Block" which performs Fast Fourier Transform on the input and gives the output in complex frequency bins. The "Complex to Mag^2"

[1] The author would like to acknowledge Ms. Jayasandhya Meenakshinathan for her Data Acquisition through RTL-SDR

block converts the complex FFT into Power Spectrum. The "Moving Average" block is used to average 100 consecutive FFT frames."Keep 1 in N" block saves only 1 spectrum out of every 100 spectrums and finally the saved spectrum is forwarded to "File Sink" block which stores the final spectrums to a file in .dat format.

4.3 Processing the Acquired Data Using Python

When the .dat file from GNU Radio was analysed using Python Libraries, it was observed that a total of 800,469 frames were captured during the 22- hour observation period. Fig (4.2) shows 1 frame out of the 800469 frames captured. Each frame represents the power received for the observed bandwidth.

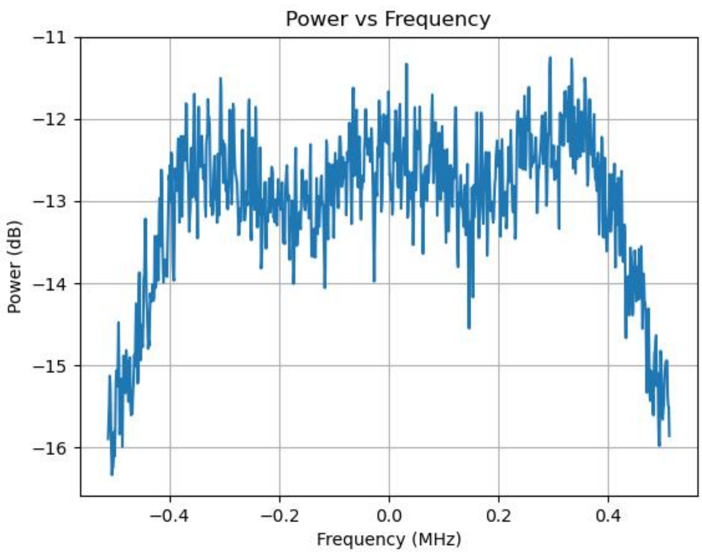

FIGURE 4.8: Power Spectrum plot

A power spectrum describes how the total power of a received signal is distributed across frequency [14]. A transmission such as the ARIES carrier at 206.5 MHz produces a sharp, concentrated peak that is higher than the noise floor. Additionally, thermal noise distributes its power uniformly across all frequency bins, producing a flat, featureless baseline that is easily distinguished from any signal of interest [15].

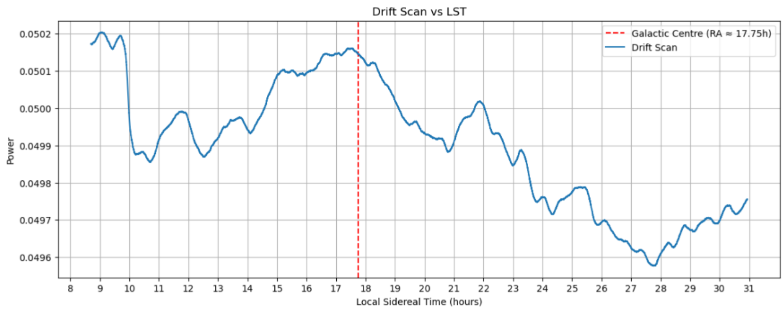

FIGURE 4.9: Drift scan plotted in Local Sidereal Time (LST). The peak near LST ≈ 17.30 h corresponds to the transit of the Galactic centre through the antenna beam.

Fig (4.9) Shows the plot of a Drift Scan. In a drift scan the antenna remains fixed while the celestial sources drift through the antenna beam pattern, due to rotation of the Earth. It was generated by plotting received power variation over time.The plot uses LST(Local Sidereal Time).LST is used in astronomy observations precisely because it directly encodes the position of the celestial sphere above the observer at any given moment [29], [35].

When a bright celestial radio source passes through the fixed antenna beam as the Earth rotates, it produces a characteristic transit peak in the drift scan. The brightest diffuse VHF radio source is the Galactic centre due to Galactic synchrotron emission [44],

If this peak occurs at the predicted time for a known source that means that the antenna beam is correctly sensitive to sky signals, the data acquisition pipeline is recording correctly, and that the data timestamps are accurate.

On 4th May 2026, the Galactic centre was expected to transit the antenna at LST ≈ 17.75h, consistent with its Right Ascension. The observed transit peak in Fig (4.9) occurs at LST ≈ 17.30h, a discrepancy of approximately 0.45h (27 minutes) earlier than predicted. One of the reasons for this could be timing error or this could be due to a pointing error [29]. The Earth rotates at a rate of 360° in 24 h

This means $15°/h$.

For an offset of 0.45 h in LST, the corresponding pointing error is:

$$\Delta\phi = 0.45 \text{ h} \times 15^\circ \text{ h}^{-1} = 6.75^\circ \tag{4.2}$$

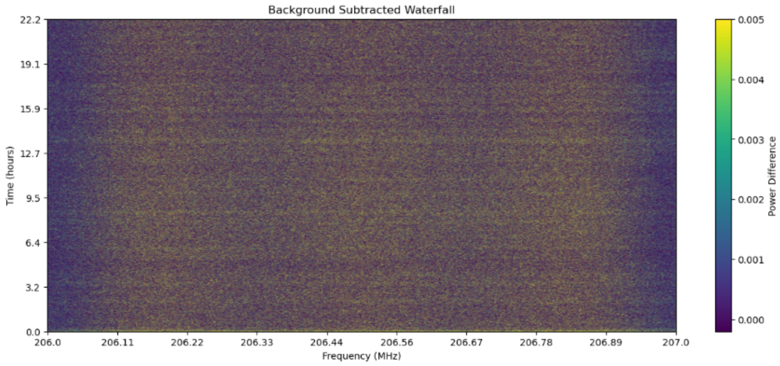

FIGURE 4.10: Noise-subtracted waterfall plot showing received power as a function of frequency and time over the 22-hour observation period

Fig (4.10) represents the variation of received power with frequency and time, after the removal of noise. The raw data that is captured contains the signal along with noise. To make sense of the signal received, it is essential to remove noise. Noise was removed to increase the visibility of weak signals.

The background subtraction was performed by computing, for each of the 1024 frequency bins, the median power value across all 800,469 frames in the dataset,and subtracting this median spectrum from every individual frame.

If a strong transient signal is present in even a small fraction of the 800,469 frames at a particular frequency, it inflates the mean at that frequency. This causes the background estimate to be too high, resulting in over subtraction that suppresses the very feature it is intended to reveal [14].For a dataset in which a genuine signal occupies far fewer frames than the noise background, the median across all frames at each frequency bin accurately represents the stable background level, unaffected by outliers in either direction [14], [10].

The absence of any clear vertical stripe at 206.5 MHz in indicates that after median subtraction the ARIES direct-path signal has been successfully removed as part of the stable background.

4.4 Multistatic Passive Radar Simulations

The bistatic simulations presented previously in this chapter demonstrate that the ARIES - IIT Kanpur geometry provides a measurable SNR for targets above a minimum detectable RCS. However, a fundamental limitation of any single bistatic pair is that a single bistatic range measurement constrains the target to a Constant Bistatic Range (CBR) ellipse but cannot determine where on that ellipse the target is located [13], [18]. This position ambiguity is the central motivation for extending the system to a multistatic configuration.
Adding a second receiver at a different location introduces two independent bistatic range measurement, producing a 2 more ellipses with two new pairs of focii [13], [33]. A target must satisfy all the range constraints simultaneously. Geometrically, this means the target must lie at the intersection of the three ellipses. [13], [18].

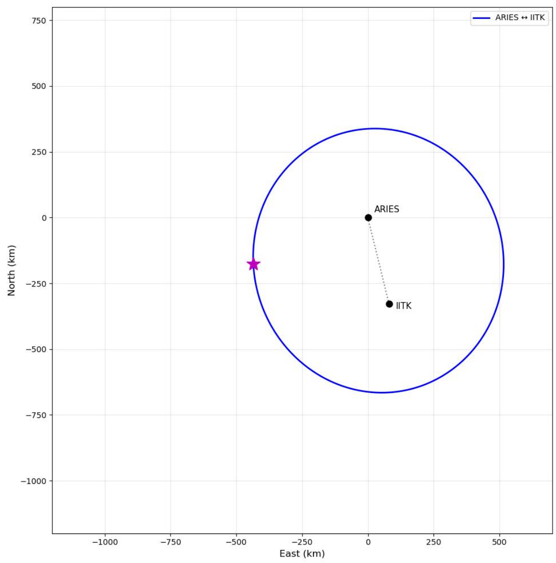

FIGURE 4.11: Target prediction in Bistatic Geometry

Fig. (4.11) shows this ambiguity for the ARIES-IIT Kanpur bistatic pair. The blue ellipse is the bistatic range sum contour corresponding to the measured bistatic range sum for the true target location marked by the magenta star. Every point on this ellipse satisfies the same range sum and is therefore an equally valid candidate target position from the perspective of a single bistatic measurement alone [13], [1]. The transmitter (ARIES) and receiver (IITK) form the two foci of the ellipse, connected by the dotted baseline. Without additional information, the system cannot resolve where on this contour the target actually lies.

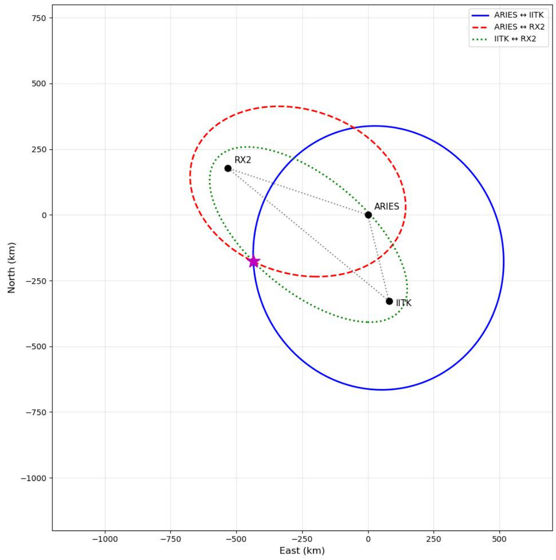

FIGURE 4.12: Target prediction in Bistatic Geometry

Adding a second receiver RX2 at a different location introduces two additional independent bistatic range measurements, one for the ARIES-RX2 pair and one for the IITK-RX2 pair.

fig(4.12) shows that each measurement produces a separate ellipse with a different pair of foci. A target must satisfy all three range constraints simultaneously, meaning it must lie at the intersection of all three ellipses. This shows that multistatic radars are better at tracking in comparison o bistatic radars.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

The above work is done to explore the possibility and feasibility of setting up a passive radar system using Yagi-Uda Antennas in Phased Arrays. First, the possibility of using FM Broadcast as Illuminators of Opportunity was explored. For this to work, a 3-element Yagi-Uda antenna was simulated and then fabricated. Experimental validation using a spectrum analyzer and Vector Network Analyzer confirmed successful reception of FM broadcast signals and acceptable antenna impedance matching characteristics. Due to very strong direct transmission Dynamic range was limited and due to this the data was not usable. The study was further expanded to other Radar Facilities in India that can act as an Illuminator of opportunity, as the FM Broadcast had very high Direct Signal. This led to using ARIES as a passive transmitter. Aries operates at 206.5 MHz and has Transmitting Power of 235 kW. The idea of using a Phased Array Receiver was explored, as that will help in improving SNR, as well as minimal detectable RCS. Simulations involving 2,3 ,4, and 7 element arrays demonstrated that increasing the number of antenna elements improved directional gain, beam steering capability, and Signal-to-Noise Ratio (SNR). Later on, a 7 element Yagi Uda Antenna was simulated and then fabricated. Long-duration signal acquisition experiments were performed using the antenna together with an RTL-SDR receiver, Low Noise Amplifier (LNA), and GNU Radio based processing chain.

5.2 Future Scope

The work presented in this thesis establishes a proof-of-concept for passive bistatic radar using the ARIES ST Radar as an illuminator of opportunity. Several directions exist for extending this work.

The phased array beam pattern simulations presented in Chapter 3 demonstrate that a 7-element array in a hexagonal geometry provides 8.45 dB of array gain over a single element, with the ability to electronically steer the beam toward any target direction. Currently, the experimental setup uses a single 7-element Yagi-Uda antenna. A natural and direct extension of this work is to physically fabricate the simulated phased array and deploy it as the receiver system.The system can then be used for aerial object detection like aircraft detection or satellite detection.

Simulations for space debris tracking have also been done. This systems can be used in space situational awareness to track space debris. The minimum RCS value can lower down further if the no. of antennas are increased.

The multistatic geometry simulations in Section 4.4 demonstrate that a second receiver completely resolves the position ambiguity of the single bistatic pair. Multiple receivers can be considered to put in variuos places to get better traking results.

Appendix A

NEC Code for 3-Element Yagi-Uda Antenna at 100 MHz

The following NEC code was used in 4NEC2 software to simulate and optimize the 3-element Yagi-Uda antenna designed for operation near 100 MHz.

```
[basicstyle=\ttfamily\small,breaklines=true]

% Appendix A

\chapter{NEC Code for 3-Element Yagi-Uda Antenna at 100 MHz}

\label{AppendixA}

\lhead{Appendix A. \emph{3-Element Yagi-Uda NEC Code}}

The following NEC code was used in 4NEC2 software to simulate and optimize the 3-element Yagi-Uda

\begin{lstlisting}[language={},basicstyle=\ttfamily\small,breaklines=true]

CM 3-el Yagi @100 MHz using your measurements (20 mm dia)
CE

' DirectorLength = 1.35 m    DirectorSpacing = +0.45 m from DE
```

```
' ReflectorLength = 1.59 m  ReflectorSpacing = -0.60 m from DE
' Driven (dipole) length = 1.425 m
' Elements along Z; boom along X; units = meters

SY rw=0.01

SY Lr=1.481064
SY Ld=1.406016
SY Ld1=1.269362

SY xr=-0.59082
SY xde=0
SY xd1=0.429054

SY hr=Lr/2
SY hd=Ld/2
SY h1=Ld1/2

' --- Geometry (41 seg each) ---

GW 1 41 xr 0 -hr   xr 0 hr   rw
GW 2 41 0  0 -hd   0  0 hd   rw
GW 3 41 xd1 0 -h1  xd1 0 h1  rw

GE 0

GN -1

EX 0 2 21 0 1 0

FR 0 1 0 0 100.0 0

RP 0 361 1 1000 90 0 1 1

EN
```

Appendix B

NEC Code for 7-Element Yagi-Uda Antenna at 206.5 MHz

```
CM 7-el Yagi @206.5 MHz
CE

SY rw=0.01
SY seg=41
SY freq=206.5

' --- Updated lengths ---

SY Lr=0.7820
SY Ld=0.6386
SY L1=0.5472
SY L2=0.5427
SY L3=0.5609
```

```
SY L4=0.4712
SY L5=0.6101

' --- Half lengths ---

SY hr=Lr/2
SY hd=Ld/2
SY h1=L1/2
SY h2=L2/2
SY h3=L3/2
SY h4=L4/2
SY h5=L5/2

' --- Updated spacings ---

SY xr=-0.47434
SY xd=0.000
SY x1=0.313712
SY x2=0.444513
SY x3=0.964822
SY x4=1.242467
SY x5=1.380598

' -------- GEOMETRY --------

GW 1 41 xr 0 -hr   xr 0 hr   rw
GW 2 41 xd 0 -hd   xd 0 hd   rw
GW 3 41 x1 0 -h1   x1 0 h1   rw
```

```
GW 4 41 x2 0 -h2   x2 0 h2   rw

GW 5 41 x3 0 -h3   x3 0 h3   rw

GW 6 41 x4 0 -h4   x4 0 h4   rw

GW 7 41 x5 0 -h5   x5 0 h5   rw


GE 0


GN -1


' Feed


EX 0 2 21 0 1 0


FR 0 1 0 0 255


RP 0 361 1 1000 0 0 1 0 90


EN
```

References

[1] M. I. Skolnik, *Radar Handbook*. New York: McGraw-Hill, 3 ed., 2008.

[2] T. Pisanu, L. Schirru, E. Urru, G. Bianchi, A. Podda, and P. Di Lizia, "The Italian BIRALET radar system to perform range and range rate measurements in the EUSST European space surveillance and tracking program," in *Proceedings of the 1st International Orbital Debris Conference*, (Sugar Land, Texas, USA), Dec. 2019.

[3] W. Orchiston and G. Swarup, "The emergence of radio astronomy in Asia: Opening a new window on the universe," in *Growth and Development of Astronomy and Astrophysics in India and the Asia-Pacific Region* (W. Orchiston, A. Sule, and M. N. Vahia, eds.), pp. 185–210, Mumbai, India: Tata Institute of Fundamental Research, 2018.

[4] S. J. Wijnholds, *Fish-Eye Observing with Phased Array Radio Telescopes*. PhD thesis, Delft University of Technology, Delft, The Netherlands, Mar. 2010.

[5] Electronics Desk, "Radiation lobes of antenna." `https://electronicsdesk.com/radiation-lobes-of-antenna.html`. Accessed: 2024.

[6] W. Kester, "MT-001: Taking the mystery out of the infamous formula, "SNR = 6.02N + 1.76 dB,"and why you should care." Analog Devices Tutorial, 2009.

[7] Everything RF, "Yagi-Uda antenna." https://cdn.everythingrf.com/community/1651683528383_637872803291478711.jpeg. Accessed: 2024.

[8] M. I. Skolnik, *Introduction to Radar Systems*. New York: McGraw-Hill, 2 ed., 1980.

[9] A. G. Westra, "Radar versus stealth: Passive radar and the future of U.S. military power," *Joint Force Quarterly*, no. 55, pp. 130–135, 2009.

[10] M. A. Richards, J. A. Scheer, and W. A. Holm, *Principles of Modern Radar: Basic Principles*. Raleigh, NC: SciTech Publishing, 2010.

[11] B. Ahuja, L. Gentile, A. Kumar, and M. Martorella, "Role of radio telescopes in space debris monitoring: Current insights and future directions," *Sensors*, vol. 25, no. 9, p. 2900, 2025.

[12] K. Jędrzejewski, M. Malanowski, M. Płotka, K. Kulpa, and M. Pożoga, "Multistatic localisation in passive radar system for LEO space objects observation using terrestrial illuminators and LOFAR radio telescope," *IET Radar, Sonar & Navigation*, 2025.

[13] H. D. Griffiths and C. J. Baker, *An Introduction to Passive Radar*. Norwood, MA: Artech House, 2017.

[14] M. A. Richards, *Fundamentals of Radar Signal Processing*. New York: McGraw-Hill, 2 ed., 2014.

[15] F. E. Nathanson, J. P. Reilly, and M. N. Cohen, *Radar Design Principles: Signal Processing and the Environment*. Raleigh, NC: SciTech Publishing, 2 ed., 1999.

[16] C. A. Balanis, *Antenna Theory: Analysis and Design*. Hoboken, NJ: John Wiley & Sons, 3 ed., 2005.

[17] T. A. Milligan, *Modern Antenna Design*. Hoboken, NJ: John Wiley & Sons, 2 ed., 2005.

[18] H. Kuschel, D. Cristallini, and K. E. Olsen, "Passive radar tutorial," *IEEE Aerospace and Electronic Systems Magazine*, vol. 34, no. 2, pp. 2–19, 2019.

[19] P. E. Howland, D. Maksimiuk, and G. Reitsma, "FM radio based bistatic radar," *IEE Proceedings – Radar, Sonar and Navigation*, vol. 152, no. 3, pp. 107–115, 2005.

[20] J. E. Palmer, B. Hennessy, M. Rutten, D. Merrett, S. Tingay, D. Kaplan, S. Tremblay, S. M. Ord, J. Morgan, and R. B. Wayth, "Surveillance of space using passive radar and the Murchison Widefield Array," in *Proceedings of the IEEE Radar Conference (RadarConf)*, (Seattle, WA), pp. 1715–1720, 2017.

[21] J. K. A. Henry and R. M. Narayanan, "Passive radar-based parameter estimation of low earth orbit debris targets," *Aerospace*, vol. 12, no. 1, p. 53, 2025.

[22] L. Creed, J. Graham, C. Jenkins, S. Diaz Riofrio, A. R. Wilson, and M. Vasile, "STRATHcube: The design of a CubeSat for space debris detection using in-orbit passive bistatic radar," in *Proceedings of the 72nd International Astronautical Congress (IAC)*, (Dubai, UAE), 2021. Paper IAC-21,A6,1,2,x66530.

[23] V. Y. Tsylov *et al.*, "Handbook of radar engineering fundamentals," tech. rep., US Army Foreign Science and Technology Center, 1970.

[24] F. Colone, F. Filippini, and D. Pastina, "Passive radar: Past, present, and future challenges," *IEEE Aerospace and Electronic Systems Magazine*, vol. 38, no. 1, pp. 4–23, 2023.

[25] J. W. A. Brown, *FM Airborne Passive Radar*. PhD thesis, University College London, London, UK, 2013.

[26] J. E. Palmer, H. A. Harms, S. J. Searle, and L. M. Davis, "DVB-T passive radar signal processing," *IEEE Transactions on Signal Processing*, vol. 61, no. 8, pp. 2116–2126, 2013.

[27] J. L. Garry, C. J. Baker, and G. E. Smith, "Evaluation of direct signal suppression for passive radar," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 7, pp. 3786–3799, 2017.

[28] D. Cristallini, I. Pisciottano, and H. Kuschel, "Multi-band passive radar imaging using satellite illumination," in *Proceedings of the IEEE International Radar Conference*, 2019.

[29] T. L. Wilson, K. Rohlfs, and S. Hüttemeister, *Tools of Radio Astronomy*. Berlin: Springer, 5 ed., 2009.

[30] J. Liu, H. Li, and B. Himed, "On the performance of the cross-correlation detector for passive radar applications," *Signal Processing*, vol. 113, pp. 32–37, 2015.

[31] C. Moscardini, D. Petri, A. Capria, M. Conti, M. Martorella, and F. Berizzi, "Batches algorithm for passive radar: A theoretical analysis," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 51, no. 2, pp. 1475–1487, 2015.

[32] S. Heunis, Y. Paichard, and M. Inggs, "Passive radar using a software-defined radio platform and open source software tools," in *Proceedings of the IEEE Radar Conference*, (Kansas City, MO), pp. 879–884, 2011.

[33] G. Fang, J. Yi, X. Wan, Y. Liu, and H. Ke, "Experimental research of multistatic passive radar with a single antenna for drone detection," *IEEE Access*, vol. 6, pp. 33542–33553, 2018.

[34] F. Colone, D. O'Hagan, P. Lombardo, and C. Baker, "A multistage processing algorithm for disturbance removal and target detection in passive bistatic radar," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 45, no. 2, pp. 698–722, 2009.

[35] ITU Radiocommunication Bureau, *Handbook on Radio Astronomy.* Geneva: International Telecommunication Union, 3 ed., 2013.

[36] H. D. Griffiths and C. J. Baker, "Passive coherent location radar systems. Part 1: Performance prediction," *IEE Proceedings – Radar, Sonar and Navigation*, vol. 152, no. 3, pp. 153–159, 2005.

[37] S. Bhattacharjee, M. Naja, A. Jaiswal, K. S. Rawat, R. Sagar, and S. Ananthakrishnan, "ARIES ST radar: The first central himalayan wind profiler," *Journal of Astronomical Instrumentation*, vol. 11, no. 4, 2022.

[38] M. Durga Rao, P. Kamaraj, J. Kamal Kumar, K. Jayaraj, K. M. V. Prasad, J. Raghavendra, T. Narayana Rao, and A. K. Patra, "The advanced indian MST radar (AIR): System description and sample observations," *Radio Science*, vol. 55, p. e2019RS006883, 2020.

[39] M. Malanowski, "Analysis of detection range of FM-based passive radar," *IET Radar, Sonar & Navigation*, vol. 8, no. 2, pp. 153–159, 2014.

[40] D. M. Pozar, *Microwave Engineering.* Hoboken, NJ: John Wiley & Sons, 4 ed., 2011.

[41] T. Zervos, A. A. Alexandridis, F. Lazarakis, K. Dangakis, and C. Soras, "Array factor measurements towards array antennas characterization," in *Proceedings of the WSEAS International Conference on Communications*, (Athens, Greece), 2007.

[42] R. D. Straw, *The ARRL Antenna Book.* Newington, CT: American Radio Relay League, 21 ed., 2007.

[43] A. Anttonen, M. Kiviranta, and M. Höyhtyä, "Space debris detection over intersatellite communication signals," *Acta Astronautica*, vol. 187, pp. 156–166, 2021.

[44] M. Remazeilles, C. Dickinson, A. J. Banday, M.-A. Bigot-Sazy, and T. Ghosh, "An improved source-subtracted and destriped 408-MHz all-sky map," *Monthly Notices of the Royal Astronomical Society*, vol. 451, no. 4, pp. 4311–4327, 2015.

[45] S. Herman and P. Moulin, "A particle filtering approach to FM-band passive radar tracking and automatic target recognition," in *Proceedings of the IEEE Aerospace Conference*, (Big Sky, MT), 2002.

[46] D. Shephard and G. Richards, "Passive target tracking using transmitters of opportunity," in *Proceedings of the Target Tracking and Data Fusion Conference*, pp. 151–157, 2008.

[47] A. Di Lallo, A. Farina, R. Fulcoli, P. Genovesi, R. Lalli, and R. Mancinelli, "Design, development and test on real data of an FM based prototypical passive radar," in *Proceedings of the IEEE Radar Conference*, (Rome, Italy), 2008.

[48] C. R. Berger, B. Demissie, J. Heckenbach, P. Willett, and S. Zhou, "Signal processing for passive radar using OFDM waveforms," *IEEE Journal of Selected Topics in Signal Processing*, vol. 4, no. 1, pp. 226–238, 2010.

[49] M. Randall, A. Delacroix, C. Ezell, E. Kelderman, S. Little, A. Loeb, E. Masson, W. A. Watters, R. Cloete, and A. White, "SkyWatch: A passive multistatic radar network for the measurement of object position and velocity," *Journal of Astronomical Instrumentation*, vol. 12, no. 1, p. 2340004, 2023.

[50] F. Savvidou, S. A. Tegos, P. D. Diamantoulakis, and G. K. Karagiannidis, "Passive radar sensing for human activity recognition: A survey," *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 5, pp. 700–706, 2024.

[51] P. Delos, B. Broughton, and J. Kraft, "Phased array antenna patterns — Part 1: Linear array beam characteristics and array factor," *Analog Dialogue*, vol. 54, no. 2, 2020.

[52] L. C. Godara, "Application of antenna arrays to mobile communications, Part II: Beam-forming and direction-of-arrival considerations," *Proceedings of the IEEE*, vol. 85, no. 8, pp. 1195–1245, 1997.

[53] Y. Gupta, "GMRT status update, new results and future plans," in *Proceedings of the 4th URSI Atlantic Radio Science Conference (AT-RASC)*, (Gran Canaria, Spain), 2024.

[54] R. J. Mailloux, *Phased Array Antenna Handbook.* Norwood, MA: Artech House, 2 ed., 2005.

[55] U. Septima, Nasrul, Firdaus, and R. Uliana, "Design and construct a yagi antenna with three reflector element to strengthen 4G signal in rural areas," *International Journal of Advanced Science Computing and Engineering*, vol. 5, no. 2, pp. 197–209, 2023.

[56] S. Haykin, *Adaptive Filter Theory.* Upper Saddle River, NJ: Prentice Hall, 4 ed., 2002.

[57] E. F. Knott, J. F. Shaeffer, and M. T. Tuley, *Radar Cross Section.* Norwood, MA: Artech House, 2 ed., 1993.

[58] S. Uda and H. Yagi, "Projector of the sharpest beam of electric waves," *Proceedings of the IRE*, vol. 16, no. 6, pp. 715–741, 1928.

[59] H. Yagi, "Beam transmission of ultra short waves," *Proceedings of the IRE*, vol. 16, no. 6, pp. 743–745, 1928.

[60] J. D. Kraus, *Antennas.* New York: McGraw-Hill, 2 ed., 1988.

[61] J. D. Kraus, *Radio Astronomy.* New York: McGraw-Hill, 1966.

[62] C. A. Balanis, *Advanced Engineering Electromagnetics.* New York: John Wiley & Sons, 1989.

[63] H. T. Friis, "Noise figures of radio receivers," *Proceedings of the IRE*, vol. 32, no. 7, pp. 419–422, 1944.

[64] D. K. Barton, *Radar System Analysis and Modeling.* Norwood, MA: Artech House, 2005.

[65] H. D. Griffiths and C. J. Baker, "Passive coherent location radar systems. Part 2: Waveform properties," *IEE Proceedings – Radar, Sonar and Navigation*, vol. 152, no. 3, pp. 160–168, 2005.

[66] C. J. Baker and H. D. Griffiths, "Bistatic and multistatic radar sensors for homeland security," in *Proceedings of the IEEE Radar Conference*, 2007.

[67] P. E. Howland, "Target tracking using television-based bistatic radar," *IEE Proceedings – Radar, Sonar and Navigation*, vol. 146, no. 3, pp. 166–174, 1999.

[68] N. J. Willis and H. D. Griffiths, *Advances in Bistatic Radar*. Raleigh, NC: SciTech Publishing, 2007.

[69] M. Cherniakov, ed., *Bistatic Radar: Emerging Technology*. Chichester, UK: John Wiley & Sons, 2008.

[70] N. J. Willis, *Bistatic Radar*. Raleigh, NC: SciTech Publishing, 2 ed., 2005.

[71] T. Tsao, M. Slamani, P. Varshney, D. Weiner, H. Schwarzlander, and S. Borek, "Ambiguity function for a bistatic radar," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 33, no. 3, pp. 1041–1051, 1997.

[72] H. D. Griffiths and N. R. W. Long, "Television-based bistatic radar," *IEE Proceedings F*, vol. 133, no. 7, pp. 649–657, 1986.

[73] J. D. Sahr and F. D. Lind, "The manastash ridge radar: A passive bistatic radar for upper atmospheric radio science," *Radio Science*, vol. 32, no. 6, pp. 2345–2358, 1997.

[74] R. Zemmari, M. Broetje, G. Battistello, and U. Nickel, "GSM passive coherent location system: Performance prediction and measurement evaluation," *IET Radar, Sonar & Navigation*, vol. 8, no. 2, pp. 94–105, 2014.

[75] D. Poullin, "Passive detection using digital broadcasters (DAB, DVB) with COFDM modulation," *IEE Proceedings – Radar, Sonar and Navigation*, vol. 152, no. 3, pp. 143–152, 2005.

[76] R. Saini and M. Cherniakov, "DTV signal ambiguity function analysis for radar application," *IEE Proceedings – Radar, Sonar and Navigation*, vol. 152, no. 3, pp. 133–142, 2005.

[77] F. Colone, P. Falcone, C. Bongioanni, and P. Lombardo, "WiFi-based passive bistatic radar: Data processing schemes and experimental results," *IEEE*

Transactions on Aerospace and Electronic Systems, vol. 48, no. 2, pp. 1061–1079, 2012.

[78] K. S. Kulpa and Z. Czekała, "Masking effect and its removal in PCL radar," vol. 152, pp. 174–178, 2005.

[79] J. Brown, K. Woodbridge, A. Stove, and S. Watts, "Air target detection using airborne passive bistatic radar," *Electronics Letters*, vol. 46, no. 20, pp. 1396–1397, 2010.

[80] M. Malanowski and K. Kulpa, "Analysis of integration gain in passive radar," in *Proceedings of the 2008 International Radar Conference*, pp. 323–328, 2008.

[81] J. E. Palmer, S. Palumbo, A. Summers, D. Merrett, S. Searle, and S. Howard, "An overview of an illuminator of opportunity passive radar research project and its signal processing research directions," *Digital Signal Processing*, vol. 21, no. 5, pp. 593–599, 2011.

[82] M. Ritchie, F. Fioranelli, and H. Borrion, "Micro UAV crime prevention: Can we help princess Leia?," in *Preventing Crime in the 21st Century* (B. L. Savona, ed.), pp. 359–376, Cham, Switzerland: Springer, 2017.

[83] X. Wan, J. Yi, Z. Zhao, and H. Ke, "Experimental research for CMMB-based passive radar under a multipath environment," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 50, no. 1, pp. 70–85, 2014.

[84] N. Levanon and E. Mozeson, *Radar Signals*. Hoboken, NJ: John Wiley & Sons, 2004.

[85] C. Laufer, *The Hobbyist's Guide to the RTL-SDR*. CreateSpace Independent Publishing Platform, 2018.

[86] S. Blackman, "Multiple hypothesis tracking for multiple target tracking," *IEEE Aerospace and Electronic Systems Magazine*, vol. 19, no. 1, pp. 5–18, 2004.

[87] R. Cardinali, F. Colone, C. Ferretti, and P. Lombardo, "Comparison of clutter and multipath cancellation techniques for passive radar," in *Proceedings of the IEEE Radar Conference*, (Boston, MA), 2007.

[88] G. Bournaka, A. Baruzzi, J. Heckenbach, and H. Kuschel, "Experimental validation of beamforming techniques for localization of moving target in passive radar," in *Proceedings of the IEEE Radar Conference (RadarCon)*, pp. 1710–1713, 2015.

[89] H. Kuschel and D. O'Hagan, "Passive radar from history to future," in *Proceedings of the 11th International Radar Symposium (IRS)*, (Vilnius, Lithuania), 2010.

[90] M. C. Jackson, "The geometry of bistatic radar systems," *IEE Proceedings F – Communications, Radar and Signal Processing*, vol. 133, no. 7, pp. 604–612, 1986.

[91] C. G. T. Haslam, C. J. Salter, H. Stoffel, and W. E. Wilson, "A 408 MHz all-sky continuum survey. II — the atlas of contour maps," *Astronomy and Astrophysics Supplement Series*, vol. 47, pp. 1–142, 1982.

[92] A. de Oliveira-Costa, M. Tegmark, B. M. Gaensler, J. Jonas, T. L. Landecker, and P. Reich, "A model of diffuse galactic radio emission from 10 MHz to 100 GHz," *Monthly Notices of the Royal Astronomical Society*, vol. 388, no. 1, pp. 247–260, 2008.

[93] J. J. Condon, "Radio sources and cosmology," *Astrophysical Journal*, vol. 188, pp. 279–286, 1974.

[94] K. M. Górski, E. Hivon, A. J. Banday, B. D. Wandelt, F. K. Hansen, M. Reinecke, and M. Bartelmann, "HEALPix: A framework for high-resolution discretization and fast analysis of data distributed on the sphere," *Astrophysical Journal*, vol. 622, pp. 759–771, 2005.

[95] C. L. Zoeller, M. C. Budge, and M. J. Moody, "Passive coherent location radar demonstration," in *Proceedings of the 34th Southeastern Symposium on System Theory*, pp. 358–362, 2002.

[96] H. D. Griffiths, "From a different perspective: Principles, practice and potential of bistatic radar," in *Proceedings of the International Radar Conference*, (Adelaide, Australia), pp. 1–7, 2003.

[97] S. Stein, "Algorithms for ambiguity function processing," *IEEE Transactions on Acoustics, Speech and Signal Processing*, vol. 29, no. 3, pp. 588–599, 1981.

[98] P. E. Howland, H. D. Griffiths, and C. J. Baker, "Passive bistatic radar systems," in *Bistatic Radar: Emerging Technology* (M. Cherniakov, ed.), pp. 247–314, Chichester, UK: John Wiley & Sons, 2008.

[99] H. Kuschel *et al.*, "Experimental passive radar systems using digital illuminators (DAB/DVB-T)," in *Proceedings of the 15th International Radar Symposium (IRS)*, 2007.

[100] A. W. Rihaczek, *Principles of High-Resolution Radar.* New York: McGraw-Hill, 1969.

[101] D. Langellotti, F. Colone, C. Bongioanni, and P. Lombardo, "Comparative study of ambiguity function evaluation algorithms for passive radar," in *Proceedings of the International Radar Symposium (IRS)*, (Hamburg, Germany), Sept. 2009.

[102] D. Petri, C. Moscardini, M. Martorella, M. Conti, A. Capria, and F. Berizzi, "Performance analysis of the batches algorithm for range-Doppler map formation in passive bistatic radar," in *Proceedings of the IET International Conference on Radar Systems*, (Glasgow, UK), Oct. 2012.